

Research on collaborative path planning of UAV swarms for urban logistics distribution in dense building environments

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ABSTRACT

This paper addresses the path planning challenges of UAV swarms in dense urban environments, where collision avoidance with high-rise buildings and Electromagnetic Interference resilience are critical. Firstly, in view of the characteristics of the urban environment, a UAV swarm path planning model is designed, which comprehensively considers the conventional constraints such as path length minimization, turning angle limitation and avoidance of buildings, as well as the collision avoidance and communication requirements within the UAV swarm and the influence of electromagnetic interference in the urban environment on the communication link, and the final objective function is given. Secondly, to effectively solve this model, a Multi-strategy Improved Wolf Pack Algorithm (MSIWPA) is proposed. An initialization strategy combining the good point set and LHS is introduced to enhance the quality of the initial population. An improved wandering strategy based on the population vitality value is designed, enabling the adaptive selection of improved search methods and better balancing the requirements of computational complexity and accuracy. Adaptive assault step sizes and encirclement step sizes are designed to effectively accelerate the convergence speed of the algorithm. A population-update strategy based on the genetic mechanism is designed to retain high-quality individuals and maintain the population at a high-quality level. Based on the above methods, simulations and comparisons of UAV swarm path planning are carried out in specific scenarios of urban environments with dense buildings. The results show that the proposed method can effectively plan the paths of UAV swarms, and it is demonstrated that MSIWPA can more effectively plan high-quality flyable paths for UAV swarms.

1. Introduction

As urbanization accelerates, both the urban population and building density are on the rise. The traditional ground-based logistics distribution model is thus confronted with issues such as traffic congestion, low distribution efficiency, and high costs. According to the World Bank's statistics in 2024, the global urbanization rate has exceeded 68 %, overburdening the traditional land-transportation system. Urban freight transportation accounts for more than 40 % of traffic-related carbon emissions, and the "last-mile" cost makes up a staggering 53 % of the total logistics cost. Particularly in areas with dense buildings, the

intricate road network and limited parking spaces impede the passage of logistics vehicles, exacerbating the "last-mile" problem in terminal distribution. In this context, UAV distribution, with its advantages of high flexibility, high efficiency, and freedom from ground-traffic constraints, has not only shown great application potential in fields like military reconnaissance (Fu & Sun, 2024) and disaster monitoring (Bashir, Ahmad, Rizvi, & El-Latif, 2024), but has also become a crucial research area for the intelligent upgrade of urban logistics. In recent years, with the rapid development of UAV technology, single-UAV delivery has been implemented in some scenarios. Nevertheless, faced with large-scale and highly dynamic urban logistics demands, the carrying capacity and coverage of a single UAV are insufficient to meet the requirements of

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Nomenclature			
$\bar{H}_b$	average height of the building	SNR_t	SNR threshold
H_{th}	Layered thresholds for flight altitude	$y_{\max}$	maximum climb rate of the UAV
H_{task}	flight altitude required by the mission	$[H_{\min}, H_{\max}]$	Flight altitude range
j	The number of the waypoint	i, v	The number of the UAV
x_{ij}, y_{ij}, h_{ij}	3D coordinates of waypoints	N	The total number of waypoints
$C_{ij,H}$	Altitude cost of waypoint	$f_{i,L}$	Path length cost of the UAV i
β	environmental airflow influencing factor	α	the obstacle influencing factor
$\theta_{ij,turn}$	turning angle of waypoint	$C_{ij,T}$	The cost of turning angle and climb angle of waypoint
$C_{ij,R}$	Obstacle avoidance cost of waypoint	$\theta_{ij, climb}$	climb angle of waypoint
D_R	the farthest distance of the simulated communication interference field	d_{ij}	Euclidean distance from waypoint j to the center of the obstacle
β_0	the path loss when the distance is 1 m in the case of line-of-sight distance	γ	The communication influencing factor
M	the number of UAVs	$d_{i,v}^s$	the distance of the distance evaluation point s of 2 UAVs in the same sequence path segment
r_{safe}	the minimum flight safety distance between UAVs to avoid collision	r_{UAV}	the radius of UAV
d_c	the path length of the signal passing inside the obstacle	$C_{j,S}$	The cost of maintaining distance between swarms of path
f_c	the frequency of the UAV transmitting signal	ω	the influencing factor that measures distance on the signal
c	the speed of light	η	the influence coefficient of the obstacle on the signal
p_t	signal transmitting power of the UAV	d	the distance between UAVs
$C_{j,I}$	The cost of communication constraints	p_n	the noise power
r	the good point	$P_n(i)$	the set of good point
H	the population vitality value	m	Dimension of the problem
$step_c^d$	the encirclement step size	$step_b^d$	the assault step size
n	the size of the population	WD	population diversity index
		$T_{\max}$	maximum number of iterations
		L_{near}	the determination distance of entering the encirclement

efficient delivery. Consequently, the collaborative delivery of UAV swarms has emerged. Through the cooperation of multiple UAVs for task assignment, route optimization, and resource scheduling, it can remarkably enhance delivery efficiency and system robustness (Li, Liu, & Jiang, 2022). UAV swarms can not only handle multiple orders in parallel but also adapt to complex urban environments through information sharing and collaborative obstacle-avoidance, such as air-flow interference between high-rises, various dynamic obstacles, and no-fly zone restrictions.

When the UAV swarm undertakes the logistics delivery task, path planning plays a critical role. Path planning is aimed at calculating the optimal flight routes for each UAV to avoid collisions, optimize resource utilization, and ensure the efficient and safe execution of the task. Precise path planning is of crucial significance in aspects such as improving the task-execution efficiency of the UAV swarm, reducing energy consumption, and ensuring flight safety. However, logistics delivery in an urban environment usually has to confront more complex constraints, which are mainly manifested in the following aspects: First, there are spatial constraints and high building density. The dense building environment in cities gives rise to a complex “urban canyon” effect. The UAV swarm needs to fly through the narrow gaps between buildings, and its path planning must take into account three-dimensional obstacle avoidance (including facade obstacles like glass curtain walls and billboards). Moreover, the wind shear and turbulence effects between tall buildings in the city have a substantial impact on the flight state of UAVs, severely affecting their stability and energy consumption. When flying between high-rises, there is also the issue of communication signals being blocked, making it difficult to maintain communication within the UAV swarm. Second, there are dynamic obstacles and airspace conflicts. In practical applications, the UAV swarm needs to adapt to a complex and changeable dynamic or static environment (Phung & Ha, 2021). Besides UAVs, there are various dynamic obstacles in the urban airspace, such as helicopters, balloons, and other logistics UAVs. These obstacles are hard to detect during environmental

modeling, yet they have a significant impact on path safety. This requires logistics UAVs to be able to adjust their flight paths in real-time to avoid collisions.

At present, traditional single-UAV or suburban path planning methods cannot effectively solve the path planning problem of UAV swarms in an urban environment. Although many achievements have been made in the research on UAV swarm path planning, such as the path planning method based on graph search, which constructs an environmental map and uses a search algorithm to find the optimal path. However, due to the high complexity of the urban environment, there is still a lack of an efficient planning method for urban-environment path planning. Existing path planning methods still have significant limitations when dealing with such complex constraints. For example, swarm-intelligence algorithms (such as PSO) have the advantage of parallel search, but premature convergence and insufficient dynamic adaptability lead to a sharp decline in the solution quality in the high-dimensional and time-varying urban environment. Reinforcement-learning-based methods rely on a large number of simulation trainings, and their policy transferability is severely restricted by the uncertainty of the urban physical field. Precise mathematical programming methods (such as MILP) struggle to meet the real-time requirements of online replanning due to the combinatorial explosion problem.

In recent years, deep learning-based methods such as deep reinforcement learning (DRL) and graph neural networks (GNNs) have shown great potential in the field of path planning. The DRL method can effectively handle high-dimensional state space by learning the optimal strategy through interaction with the environment. GNNs are good at capturing structured information about the environment, such as the topological relationships of road networks or building complexes. However, these learning methods often face common challenges: first, they rely heavily on large amounts of high-quality training data or time-consuming simulation interactions, which are expensive to train; second, the generalization ability of its strategy is limited by the distribution of the training environment, and the performance may be

significantly reduced in the face of a new and highly dynamic urban environment. Third, the precise integration of complex physical constraints (such as UAV dynamics, swarm collision avoidance) into the learning process is still an open issue. Therefore, well-designed meta-heuristics still have irreplaceable value in online planning scenarios that require high reliability, strong generalization capabilities, and rapid deployment without pre-training. Compared with the “black box” learning method, the meta-heuristic algorithm is model-free and does not depend on a specific environmental model. It is free to train without the need for an offline training phase, thus ensuring its plug-and-play ability to unknown environments.

As a swarm intelligence algorithm that simulates the predation behavior of wolves, the wolf pack algorithm (WPA) achieves a dynamic balance between global search and local exploitation by imitating the social hierarchy of wolf packs (including the lead wolf, scout wolves, and assault wolves) and their predation behaviors (wandering, summoning, and besieging) (Wu, Zhang, & Wu, 2013). Many scholars choose the wolf pack algorithm when attempting to solve NP-hard problems such as combination optimization, task scheduling, and set partitioning. Based on the above considerations, this paper proposes a Multi-Strategy Improved Wolf Pack Algorithm (MSIWPA) that incorporates a wandering strategy based on the population vitality value and a population renewal mechanism based on the genetic strategy. Meanwhile, the adaptive assault step size is designed, effectively enhancing the performance of the algorithm. The main contributions of this paper are as follows:

- (1) An Enhanced Optimization Framework: We introduce a suite of strategic improvements to the WPA, transforming it into a more adaptive and powerful optimizer. The novelty of this framework lies in its integrated approach, which includes a population vitality metric for dynamic search strategy selection and a genetic-inspired population renewal mechanism. These enhancements collectively address fundamental limitations of classic swarm intelligence algorithms, such as premature convergence and poor dynamic adaptability, by fostering a more effective balance between global exploration and local refinement.
- (2) A High-Fidelity Planning Model for Urban Canyons: Moving beyond conventional models, we formulate a comprehensive path planning objective function that explicitly encapsulates the multifaceted constraints of dense urban operations. This model holistically integrates 3D obstacle avoidance, intra-swarm collision and communication maintenance, and resilience to electromagnetic interference, providing a realistic computational basis for safe and efficient cooperative flight.
- (3) A Rigorous Empirical Validation: We substantiate the efficacy of our proposed MSIWPA through comparative simulations in a meticulously constructed 3D urban environment that mirrors the complexities of the “urban canyon” effect. The results provide compelling evidence that our algorithm achieves superior performance in planning high-quality, feasible paths for UAV swarms compared to other state-of-the-art methods.

The main arrangement of this paper is as follows: The second section describes the UAV swarm path planning problem. In the third section, a UAV swarm trajectory planning model with necessary constraints is proposed, and then the corresponding objective function is designed. Section 4 proposes a multi-strategy improved wolf pack algorithm. In the fifth section, the improved algorithm is applied to the UAV swarm path planning problem to verify its effectiveness. Section 6 summarizes the work of this paper and its future development direction.

2. Related work

The collaborative path planning problem for UAV swarms involves selecting a safe, feasible, and efficient path from the starting point to the

destination for UAV swarms under specific constraints when they are performing various complex tasks. This is a typical optimization problem, aiming to minimize the path length while satisfying multiple constraints. In an environment with dense buildings, UAVs need to start from different starting points and fly to the same target point. The buildings and other obstacles create a complex flight environment. The goal of path planning is to find the shortest and feasible path for each UAV, so as to minimize the path cost of the entire UAV swarm and meet all the constraint conditions, as shown in Fig. 1.

During the path planning process, the following key constraint conditions need to be considered.

- (1) Path Smoothness. The generated path must be smooth enough to meet the maneuverability requirements of UAVs.
- (2) Obstacle Avoidance. Each UAV must avoid colliding with buildings and other obstacles to ensure flight safety.
- (3) Anti-collision among UAVs. During the flight, a sufficient safe distance must be maintained between any two UAVs to avoid mid-air collisions.
- (4) Maintenance of Communication Connection. Throughout the entire flight, UAVs must maintain a stable communication connection to ensure the real-time sharing of information and the effectiveness of collaborative decision-making. Considering the occlusion and interference of communication signals in the dense building environment, this constraint condition is particularly crucial.

In addition, energy consumption is also a key factor that must be taken into account during the path planning process. According to the Generalized Propulsion Energy Consumption Model (PECM) for rotary-wing UAVs proposed by Dai et al. (Ait-Saadi et al., 2023), the energy consumption level of UAVs is closely related to factors such as flight altitude, horizontal turns, and path length. Newton’s laws of motion and the principles of aerodynamics reveal that when a UAV climbs, it requires additional thrust to counteract gravity, and turning maneuvers require lateral forces and greater thrust support. Even in a stable flight state, a UAV needs to continuously do work to resist air resistance, thus creating a continuous demand for power. It can be seen that flight altitude, horizontal turns, and path length have a significant impact on energy consumption. Therefore, during the path planning process, it is necessary to comprehensively balance the relationship between energy consumption and path length to seek an optimal equilibrium.

UAV path planning is regarded as a difficult NP optimization problem (Peng et al., 2025). Over the past few decades, many scholars have been committed to finding path planning methods with better performance, which can be divided into four categories according to the algorithms used: sample-based methods, graph-oriented algorithms, mathematical planning methods, and artificial intelligence methods. Sample-based methods explore the environment through random sampling and construct feasible paths. These include the Probabilistic RoadMap (PRM) algorithm (Tu, Deng, Li, Song, & Zheng, 2024) and the Rapidly-exploring Random Tree (RRT) (Kříz & Vonásek, 2025). Although these methods do not need to know the detailed information of the entire environment in advance, they usually can only find feasible solutions rather than optimal solutions. Graph-oriented algorithms abstract the environment into a graph structure and find the optimal path by searching the graph. Examples include the A* algorithm (Jinyang, Yuhua, Ruixuan, Hao, & Kejing, 2022), Dijkstra algorithm (Chi, Ding, Wang, Chen, & Sun, 2021), Voronoi Diagram (VD) algorithm (Welzl, 1985), and Visibility Graph (VG) algorithm (Wirth, Oettershagen, Ambuhl, & Siegwart, 2015). Although these algorithms can obtain optimal solutions, as the dimension increases, the number of nodes and edges in the graph grows exponentially, making the calculation extremely difficult and unable to meet the requirements of practical applications. Mathematical planning methods describe the constraint conditions and objective functions in path planning by establishing

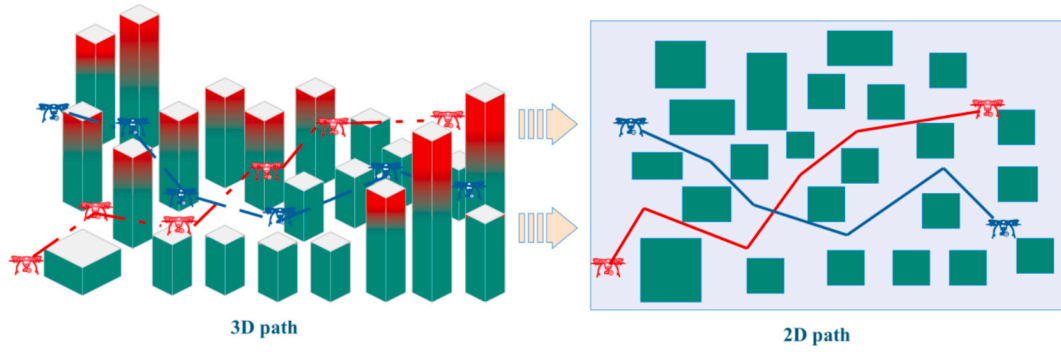


Fig. 1. Schematic diagram of UAV swarm path planning.

precise mathematical models, including Dynamic Programming (DP) (Zhang, Zhang, Tang, & Chen, 2023), Mixed-Integer Linear Programming (MILP) (Lin, Gao, Wu, & Suganthan, 2023), etc. This method has difficulties in solving non-linear optimization problems, has poor real-time performance, and the calculation time will increase significantly as the scale of the problem increases.

The last category is artificial intelligence methods, including reinforcement learning (Cao, Zhao, Wang, Xiang, & Sartoretti, 2024; Vashisth, Rückin, Magistri, Stachniss, & Popović, 2024), heuristic algorithms, meta-heuristic algorithms, etc. Among them, meta-heuristic algorithms are widely applied to the UAV path planning problem, mainly consisting of four types: swarm intelligence algorithms (SI), physics-based algorithms (Kuok et al., 2024), evolutionary algorithms (Zhou, Zhou, Deng, Yin, & Zhang, 2023), and human-based algorithms (Liu et al., 2023). Swarm intelligence algorithms are a major component of them. Their principle is to simulate the intelligent behaviors of biological groups in nature, such as Ant Colony Optimization (ACO) (Comert & Yazgan, 2023) and Artificial Bee Colony (ABC) (Mishra et al., 2020). When solving problems, these algorithms do not require the specific structure of the problem nor rely on interactions with the environment. Even if the optimal solution is not found in a short time, a sub-optimal solution can be obtained, which is often acceptable in practice (Wang, Fu, Han, Xing, & Wu, 2024). Therefore, they are widely used to solve path planning problems. For example, some scholars (Chen, Zhao, & Wang, 2021) proposed an improved sparrow algorithm that combines the uniformity metric WD to improve the quality of the initial population, and designed a boundary update strategy based on population uniformity, which greatly enhances the global search ability of the algorithm. Experimental results show that the improved algorithm exhibits better search accuracy and speed than other algorithms in 2D and 3D environments. Ref. (Xiao, Dong, & He, 2025) proposed an adaptive particle swarm optimization algorithm (APSO) to solve the UAV path planning problem, which adaptively adjusts the inertia weight and two learning factors.

In Ref. (Xu, Li, Zhou, Mao, & Huang, 2022), an incremental conductance method based on the hybrid global wolf pack algorithm (GWPA-FINC) was proposed. It was compared with multiple swarm intelligence algorithms in seven scenarios, demonstrating that the proposed GWPA-FINC effectively addresses the GMPP optimization challenge. In Ref. (Peng, Zhan, Wu, Wang, & Lai, 2025), a multi-discrete wolf pack algorithm (MDWPA) was put forward to solve the task allocation problem of UAVs in complex dynamic environments. However, although the wolf pack algorithm has shown certain advantages in solving the UAV path planning problem, like most swarm intelligence algorithms, its performance is influenced by various factors, such as the initial population and parameter settings. To improve the algorithm's solution accuracy and broaden its application scope, many scholars have attempted to propose numerous improved versions from perspectives such as adaptive parameters, population initialization, update strategies, and integration with other algorithms. In Ref. (Jiang, Yu, Han,

Chen, & Li, 2024), the selection, crossover, and mutation of the Differential Evolution (DE) algorithm were integrated to enhance the randomness of the wandering behavior, and multiple encirclement strategies were designed to strengthen the global search ability. In Ref. (Peng, Wu, Li, & Wang, 2024), inspired by the division of labor in wolf packs, an alliance formation mechanism based on individual democratic choice and centralized group decision-making was designed, aiming to solve the dynamic allocation problem of UAV swarms.

3. UAV path planning model

3.1. Hierarchical planning strategy

The environment with dense buildings usually features tall buildings, narrow streets, and limited flight space. These characteristics pose challenges to the flight path planning of UAVs, as UAVs need to avoid obstacles in a complex three-dimensional space while maintaining an efficient flight route. Among them, space fragmentation is a significant issue. Observing the real-world urban environment, it can be found that within a certain range, tall buildings are concentrated in one continuous area, while low-rise and small buildings are usually concentrated in another continuous area, and it is rare for tall and low-rise buildings to be interlaced. The height and density of buildings cause the flight space to be divided into multiple non-continuous regions, thus presenting a fragmented distribution. UAV flights are affected by air currents. Buildings in the low-altitude space can block part of the air currents and reduce some air resistance; although the air currents have a greater impact at high altitudes, there are fewer obstacles. Therefore, simple linear height-cost calculation is too simplistic and does not conform to the actual situation. To address this challenge, a hierarchical planning strategy is introduced. To achieve the three-dimensional space division in the environment with dense buildings, factors such as building height, UAV flight characteristics, and safety constraints need to be comprehensively considered. The following is a division model and equations based on multi-factor decision-making.

(1) Calculation of the Layered threshold

In the three-dimensional space planning of UAVs in a dense-building environment, the layered threshold is the height baseline for dividing the flight levels of UAVs, which determines the boundaries of different altitude layers. By quantifying the layered threshold, the continuous three-dimensional space can be discretized into a low-altitude zone that is close to the ground and requires frequent obstacle avoidance, and a high-altitude zone with straighter paths but is more significantly affected by air currents. The calculation equation is:

$$H_{th} = \bar{H}_b \times (0.6 + \frac{y_{max}}{10}) + \delta \quad (1)$$

where $\bar{H}_b$ is the average height of the building; y_{max} is the maximum

climb rate of the UAV, which is determined by the average speed and maximum elevation angle of the UAV between buildings; δ is the safety margin set to deal with bursts of airflow or positioning errors.

(2) Calculation of the optimal height layer

Depending on the different tasks that UAVs perform, their requirements for flight altitude also vary. For example, a reconnaissance mission may require a higher flight altitude to obtain a better field of view, while a delivery mission can reduce the flight altitude. Therefore, after obtaining the layered threshold, by integrating the actual task requirements, the requirements of different tasks for the UAV flight altitude are abstracted into an optimal flight altitude interval $[H_{\min}, H_{\max}]$, as shown in Fig. 2. The calculation equation is as follows:

$$H_{\min} = H_{th} - H_{task1}; H_{\max} = H_{th} + H_{task2} \quad (2)$$

where H_{task} and H_{task2} are the flight altitudes determined according to the task requirements. These two values are not necessarily equal, and their values are completely determined by the task requirements.

The core concept of a layered planning strategy is to partition the flight space into two levels and compute the path costs separately for each level. This approach helps to simplify the path planning problem, boost planning efficiency, and guarantee the flight safety of UAVs in complex environments. Through layered planning, UAVs can pick the optimal path within different altitude levels. This enables them to avoid frequently evading buildings during low-altitude flights, thus enhancing flight efficiency. Moreover, by means of layered planning, UAVs can pre-plan their flight paths at different altitude levels, mitigating the impact of sudden obstacles on their flights. Additionally, it helps to reduce communication interference among UAVs, improve communication quality, and thereby strengthen the robustness of UAV swarm cooperation.

3.2. Path representation

The UAV kinematics model adopted in this study is based on the assumption that the UAV is a particle model with full six-degree-of-freedom motion capabilities (pitch, yaw, roll) and is constrained by the maximum turning angle and the maximum climb angle. These parameters directly reflect the maneuverability limitations of real UAVs. The waypoint is generated taking into account the actual flight characteristics of the UAV, including inertia effects and continuous smooth trajectory requirements. In this paper, it is assumed that the flight speed of UAV is fixed. Each UAV path is discretized into N waypoints (including the starting point and the ending point). The coordinates of the waypoint j of the UAV i are denoted as x_{ij}, y_{ij}, h_{ij} . Through the discrete representation of the path, a B-spline curve is used to smooth the trajectory, obtaining a complete UAV path that meets practical requirements. However, when planning the paths of a UAV swarm, multiple constraint conditions need to be considered. Even if the discrete waypoints are in appropriate positions, the path segments between them may not necessarily satisfy the constraint conditions. Generally speaking, the larger N is, that is, the more detailed the path division is,

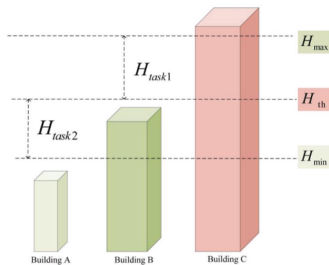


Fig. 2. Schematic diagram of a layered planning strategy.

the smaller the path segments between the waypoints will be. Provided that all waypoints satisfy the constraint conditions, the probability that all path segments satisfy the constraint conditions will increase. Of course, the corresponding consumption of computing resources will be greater, and the calculation time will also be longer. Therefore, the number N of waypoints should be adjusted appropriately according to specific practical needs.

3.3. Objective function

The ultimate solution goal of UAV path planning among dense buildings is to find a feasible flight route with the minimum path length for the entire UAV swarm. Of course, the optimal route for the entire UAV swarm is not necessarily composed of the optimal routes of each individual UAV. The key to its “feasibility” lies in the fact that it must satisfy five constraint conditions under the given circumstances, namely, path height constraint, turning-angle constraint, obstacle-collision-avoidance constraint, collision-avoidance constraint among UAVs, and communication-maintenance constraint. Based on the above-mentioned constraint conditions and the characteristics of the dense-building environment, this paper designs an objective function that synthesizes height cost, turning-angle cost, obstacle-avoidance cost, swarm-collision-avoidance cost, communication-maintenance cost, and path-length cost. The calculation equations and models for each of these costs and the path cost are as follows:

(1) Path length cost

In dense building environments, the path cost function in the objective function of UAV swarm path planning is the core component, which mainly measures the geometric characteristics of the path. The total path length is the sum of the Euclidean distances between all adjacent waypoints, which can be minimized to reduce flight time and energy consumption. The cost function associated with the path length can be expressed as follows:

$$f_i^L = 1 - \frac{\sqrt{(x_{iN} - x_{i1})^2 + (y_{iN} - y_{i1})^2 + (h_{iN} - h_{i1})^2}}{\sum_{j=2}^N \sqrt{(x_{ij} - x_{i,j-1})^2 + (y_{ij} - y_{i,j-1})^2 + (h_{ij} - h_{i,j-1})^2}} \quad (3)$$

where the denominator calculates the actual path length of UAV i , which is the sum of the Euclidean distances between all adjacent waypoints; the numerator calculates the straight-line distance from the starting point to the ending point of UAV i . Obviously, the value of f_i^L is in the range of 0 to 1. The smaller the value of f_i^L , the lower the cost of the corresponding actual path length. The proposed path planning algorithm is designed to identify the path with the minimum length cost. However, it is not calculated directly through the above equation, and the path costs brought by the five constraint conditions mentioned above also need to be considered. In reality, a short path length does not necessarily correspond to the optimal path. A “shortcut” often means taking greater risks, such as having to fly close to buildings or making larger turning angles. Therefore, in this paper, a penalty mechanism based on constraint conditions is designed to quantify the “risk” of each path, and then corresponding weights are assigned to balance the path-length cost and the path risk. The specific quantification equations for each constraint condition will be introduced in the following text.

(2) Altitude cost

In reality, during flight, UAVs often need to continuously adjust their flight altitude, overcome gravity and air resistance, resulting in greater energy consumption. Considering the actual situation of dense urban buildings and mission requirements, flying at too low an altitude may not only face more obstacles and cause disturbance to residents, but also

fail to meet mission requirements in some mission scenarios (such as reconnaissance missions). Flying at too high an altitude may encounter more complex atmospheric conditions and air resistance, leading to increased energy consumption. Given the complexity of the urban dense-building environment, this paper abstracts the UAV flight space into an optimal flight interval. Therefore, different equations are used to calculate the altitude cost at different altitude levels, which is more in line with the actual situation and makes the path planning more accurate. The calculation equation for the height cost of each waypoint is as follows:

$$C_{ij}^H = \begin{cases} e^{\frac{H_{\min}-h_{ij}}{H_{\min}}}, & h_{ij} < H_{\min} \\ 0, & H_{\min} \leq h_{ij} \leq H_{\max} \\ \left(\beta \times \frac{h_{ij}-H_{\max}}{H_{\max}}\right)^2, & h_{ij} > H_{\max} \end{cases} \quad (4)$$

where h_{ij} is the flight altitude of UAV i at waypoint j , $H_{\min}$ and $H_{\max}$ represent the upper and lower limits of the optimal flight interval of the UAV's current path, α is the obstacle influencing factor, and β is the environmental airflow influencing factor.

(3) The cost of turning angle and climb angle

UAVs also consume energy when adjusting their flight angles through maneuvers. The smoother the path, the better it's stability. When planning a path, paths with smaller turning angles should be selected as much as possible. In this paper, the stability of the path is measured by the magnitude of the turning angle and climbing angle as well as the frequency of rotation. The calculation equation for the cost of the turning angle and the climbing angle at the waypoint j of the path i is as follows:

$$C_{ij}^T = \begin{cases} +\infty, & \theta_{ij,turn} > \theta_{turnmax} \text{ or } \theta_{ij,climb} > \theta_{climbmax} \\ \frac{\theta_{ij,turn}}{\theta_{turnmax}} + \frac{\theta_{ij,climb}}{\theta_{climbmax}}, & \text{else} \end{cases} \quad (5)$$

where $\theta_{turnmax}$ and $\theta_{climbmax}$ are the maximum turning angle and the maximum climb angle of the UAV. The setting of turn angle constraints and climb angle constraints is based on the actual maneuverability of a typical multi-rotor UAV. The maximum turning angle ensures that the UAV can safely avoid obstacles in dense built environments while avoiding instability due to overload. The climb angle constraint ensures that the vertical speed and acceleration of the UAV are within the allowable range of the power system. The equation for calculating the turn angle $\theta_{ij,turn}$ and climb angle $\theta_{ij,climb}$ of waypoint j is as follows:

$$\theta_{ij,turn} = \arctan\left(\frac{\|(x_j - x_{j-1}, y_j - y_{j-1}) \times (x_{j+1} - x_j, y_{j+1} - y_j)\|}{(x_j - x_{j-1}, y_j - y_{j-1}) \cdot (x_{j+1} - x_j, y_{j+1} - y_j)}\right) \quad (6)$$

$$\theta_{ij,climb} = \left\| \arctan\left(\frac{h_j - h_{j-1}}{\sqrt{(x_j - x_{j-1})^2 + (y_j - y_{j-1})^2}}\right) - \arctan\left(\frac{h_{j+1} - h_j}{\sqrt{(x_{j+1} - x_j)^2 + (y_{j+1} - y_j)^2}}\right) \right\|$$

(4) Obstacle avoidance cost

In view of the characteristics of the urban environment with dense buildings, UAVs need to frequently avoid building obstacles during flight. This is a necessary constraint. Once a collision occurs, the path

cost is set to infinity. Besides, there are a large number of civilian electromagnetic signals in the urban environment, which cause certain interference to the UAV's information and communication links. In this paper, such communication interference is abstracted as a communication interference field centered on the building and gradually weakening with the increase of distance. Each waypoint is regarded as a risk assessment point, and a total of $N - 2$ waypoints are evaluated (excluding the starting point and the ending point). The specific calculation equation for the obstacle avoidance cost at the waypoint j of path i is as follows:

$$C_{ij}^R = \begin{cases} +\infty, & x_{\min} < x_{ij} < x_{\max} \& y_{\min} < y_{ij} < y_{\max} \& R_{\min} < h_{ij} < R_{\max} \\ \gamma \times (1 - e^{d_{ij}/D_R}), & d_{ij} < D_R \\ 0, & d_{ij} > D_R \end{cases} \quad (7)$$

where $x_{\min}, x_{\max}, y_{\min}, y_{\max}, R_{\min}$ and $R_{\max}$ are used to describe the horizontal range and height range of the obstacle, forming a cuboid space. d_{ij} is the Euclidean distance from the waypoint j to the center of the obstacle, D_R is the farthest distance of the simulated communication interference field, and γ is the communication influencing factor.

(5) The cost of maintaining distance between swarms

For the path planning of UAV swarms, it is very important to avoid collisions between swarms. In order to facilitate the calculation, this paper does not consider the time difference between each UAV to each path node, that is, it is assumed that each UAV reaches the j^{th} waypoint at the same time. The path segment between the j^{th} waypoint and the $j + 1^{th}$ waypoint is also evenly divided into N distance evaluation points, and the distance between the s^{th} distance evaluation point of UAV i and UAV v is required to be greater than the safe distance. Set the cost of keeping the distance of the path that violates the constraint to infinity, otherwise it will be written as 0, as shown in the following equation:

$$C_j^S = \begin{cases} +\infty, & d_{i,v}^s < 2r_{UAV} + r_{safe} \\ 0, & d_{i,v}^s > 2r_{UAV} + r_{safe} \end{cases}; (i, v \in [1, M]; s \in [1, N]) \quad (8)$$

where $d_{i,v}^s$ is the distance of the distance evaluation point s of UAV i and UAV v in the same sequence path segment, M is the number of UAVs in the swarm, r_{UAV} is the radius of UAV, and r_{safe} is the minimum flight safety distance between UAVs to avoid collision.

(6) The cost of communication constraints

The channel model and SNR calculation follow the established air-to-air communication model for UAVs, which is widely adopted in UAV

swarm research (Bai, Li, Fang, Zhang, & Tao, 2020; Yan, Fu, Zhang, & Wang, 2019). UAV swarms need to maintain communication at all times during flight, requiring each UAV to be able to communicate with any other UAV. The communication between UAVs can be directly established between the two, or relay communication can be carried out through other UAV nodes, so to judge whether the entire UAV swarm meets the communication requirements, it is only necessary to randomly judge whether one UAV can communicate with all other UAVs. At each

moment, there are two situations between UAV i and UAV v : an obstacle-free line-of-sight environment (LoS) and an obstacle-blocked non-line-of-sight environment (NLoS). In both cases, the channel coefficient of the waypoint j is calculated separately (Zhu et al., 2024):

$$\beta(d) = \begin{cases} \beta_0 \times d^{-\omega}, & \text{LoS environment,} \\ k \times \beta_0 \times d^{-\omega}, & \text{NLoS environment,} \end{cases} \quad (9)$$

$$\beta_0 = \left(\frac{c}{4\pi \times f_c}\right)^2, \quad k = 10^{\frac{-10}{\eta \times d_c}}$$

where β_0 is the path loss when the distance is 1 m in the case of line-of-sight distance, d is the distance between UAVs at waypoint j , c is the speed of light, ω is the influencing factor that measures distance on the signal, f_c is the frequency of the UAV transmitting signal, η is the influence coefficient of the obstacle on the signal, affected by its material, temperature and other factors, d_c is the path length of the signal passing inside the obstacle. The signal-to-noise ratio (SNR) is further calculated as follows:

$$\text{SNR} = \beta(d) \times 10^{\frac{p_t - p_n}{10}} \quad (10)$$

where p_t is the signal transmitting power of the UAV, and p_n is the noise power. When the signal-to-noise ratio is greater than the threshold, it is considered that there is an information link between the UAVs and communication can be carried out. Once the signal-to-noise ratio is less than the threshold, it is believed that the two UAV nodes cannot communicate, and the cost of this path is set to infinity. In summary, the steps to calculate the communication constraint cost of a path are as follows: First, traverse each waypoint. For each waypoint, randomly select a UAV and calculate according to Eq. (10) above to determine whether there is a situation where the SNR between two UAVs is less than the threshold. If so, the communication constraint cost of this path is infinity; otherwise, it is 0, as shown in the following equation:

$$C_j^t = \begin{cases} +\infty, & \text{SNR}_{i,v}^j < \text{SNR}_t \\ 0, & \text{SNR}_{i,v}^j \geq \text{SNR}_t \end{cases} \quad (11)$$

(7) Overall objective function

As described above, after synthesizing the five constraint costs, the calculation equation of the overall objective function of the UAV swarm planning path is as follows:

$$F_{obj} = \sum_{i=1}^M (f_i^L + \sum_{j=2}^{N-1} (C_{ij}^H + C_{ij}^T + C_{ij}^R)) + \sum_{j=1}^{N-1} (C_j^S + C_j^t) \quad (12)$$

Table 1 systematically lists all path planning constraints, their mathematical formulas, and brief descriptions.

Table 1
Constraint summary table.

Constraint category	Equations	Description
f_i^L	Eq. (3)	Path length cost for the UAV i
C_{ij}^H	Eq. (4)	Altitude cost for the UAV i at the waypoint j
C_{ij}^T	Eq. (5)	Turning & climb angle cost for the UAV i at the waypoint j
C_{ij}^R	Eq. (7)	Obstacle avoidance & communication interference cost for the UAV i at the waypoint j
C_j^S	Eq. (8)	Inter-UAV collision avoidance cost for the swarm at the path j segment
C_j^t	Eq. (11)	Communication maintenance cost for the swarm at the waypoint j

4. Multi-strategy improved wolf pack algorithm

This chapter details the components of the proposed MSIWPA, which synthesizes four key enhancements to the conventional WPA to address the challenges of UAV swarm path planning in dense urban environments. The MSIWPA framework incorporates: (1) Population initialization strategy based on the Good Point Set and Latin Hypercube Sampling (LHS); (2) Wandering behavior based on population vitality value, which dynamically selects from four distinct search strategies to balance exploration and exploitation; (3) Adaptive step size strategy to accelerate convergence; (4) a Population renewal mechanism based on genetic effects.

4.1. Population initialization strategy based on the good point set and LHS

Population initialization is the first step in the operation of swarm intelligence algorithms. Although different swarm intelligence algorithms adopt various initialization strategies, their core idea is to generate the population randomly. The wolf pack algorithm also uses a random generation method when initializing the population. The core advantage of this initialization method lies in its simplicity and ease of implementation, as well as a certain global exploration ability. However, the disadvantage is that it may lead to uneven distribution of populations in the solution space, so that some regions may be oversampled while others may be ignored, and may not provide a good enough initial solution for the algorithm, resulting in more iterations required to converge to the optimal solution. Research in relevant literature shows that the quality of the initial population has a significant impact on the probability and speed of finding the global optimal solution. Improving the uniformity of the initial population can enhance the diversity of the population and accelerate the convergence speed (Wang et al., 2024). Currently, many scholars have adopted different strategies to improve population initialization (Wu & Xiao, 2020). The good point set is a deterministic sampling method based on number theory, aiming to cover the solution space by selecting sample points with a uniform distribution. The good point set has the characteristic of low-deviation, which can ensure that the sample points are evenly distributed in the solution space. It was initially proposed by Li, Chen, and Shi (2023), and its basic definition and construction are as follows.

Assuming that G_m is a Euclidean geometric space in the m dimension, if $r \in G_m$, then $P_n(i) = (r_1 \times i_1, r_2 \times i_2, \dots, r_n \times i_n)$, $i = 1, 2, \dots, n$, where n represents the number of samples, $P_n(i)$ represents the set of good point, and r refers to the good point, this paper randomly selects the equation for generating the good point, as follows:

$$\begin{cases} r = \left\{ 2\cos\left(\frac{2\pi \times j}{k}\right) \times i, 1 \leq i \leq n, 1 \leq j \leq m \right\} \text{rand} \leq 0.5 \\ r = \{e^j \times i\} \text{rand} > 0.5 \end{cases} \quad (13)$$

where k is the smallest prime that satisfies $(k-3)/2 \geq m$. The steps to construct an initial population from a good spot set are as follows.

Step 1: Calculate the value of $r.r = r_1, r_2, \dots, r_n$ can be calculated according to Eq. (13).

Step 2: Construct a good point set P_n with a quantity of n .

Step 3: Map the good point set P_n onto the value space domain where the population is located.

$$P_n(i) = (r_1 \times i_1, r_2 \times i_2, \dots, r_n \times i_n); (i = 1, 2, \dots, n) \quad (14)$$

$$X_i^j = lb_j + P_n(i) \times (ub_j - lb_j) \quad (15)$$

where ub_j represents the upper limit of the value in the current dimension, and lb_j represents the lower limit of the value in the current dimension, X_i^j represents the coordinate of the individual i in the dimension j . Based on the excellent uniformity of the good point set

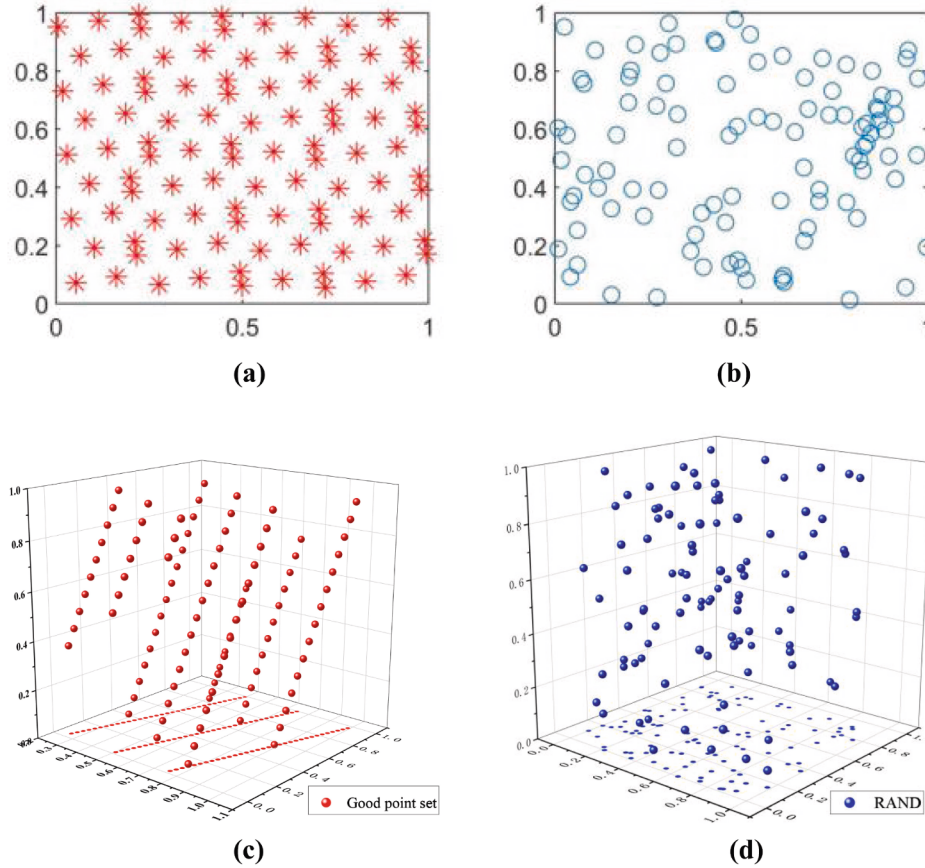


Fig. 3. Comparison of the location distribution of populations (dimension = 2,50).

method, this paper applies it to the population initialization process to improve the quality of the initial population. Fig. 3 shows the position distribution of the initial populations obtained by the good point set method and the random method, as follows.

Fig. 3 consists of four sub-figures. Sub-figure (a) and (b) are schematic diagrams of the population positions generated by the good point set method and the random method respectively in a 2-dimensional space. Sub-figure (c) and d are schematic diagrams of the population positions generated by the good point set method and the random method respectively in a 50-dimensional space. It can be seen that in the low-dimensional space, the population generated by the good point set method has a significantly better uniform distribution. The population generated by the good point set method shows a relatively regular grid-like distribution, while the randomly initialized population shows obvious aggregation. This means that in the iterative process, the initial-generation solutions of the good point set population are more likely to be close to the global optimal solution, enabling the algorithm to find the optimal value more quickly and significantly reducing the number of iterations of the algorithm.

However, in the experiments, we found that in medium-and high-dimensional spaces, although the population generated by the good point set is still more uniformly distributed than the randomly generated population, there are still some aggregation areas. Moreover, as the dimension continuously increases, the performance of the good point set method significantly declines. In sub-figure (c) and (d), the population positions generated by the good set method and the random method are also shown, with the dimension increased to 50. It can be seen from the figure that when the dimension is 50, the position distribution of the population in sub-figure (c) is band-shaped, leaving a large blank area in the middle, reflecting the decline in the performance of the good point set method. Compared with the randomized population in sub-figure(d),

it has lost its advantage. At the same time, in high-dimensional spaces, since more sample points need to be generated to cover the solution space, the computational cost may increase significantly. After in-depth analysis of the limitations of the good point set method in high-dimensional spaces, this paper proposes a method that combines the good point set method with LHS.

LHS is a stratified sampling method. It divides each dimension of the solution space into several intervals and randomly selects a sample point within each interval. The sample points generated by it are evenly distributed and have strong coverage, making it suitable for high-dimensional solutions. Its implementation process is as follows:

There are m dimensions, and the intervals of each dimension are divided into n equal probability sub-intervals, so that $i = 1, 2, \dots, n$ represents the sample number and $j = 1, 2, \dots, m$ represents the dimension. For each dimension j :

- (1) First, a permutation π_j is randomly generated, so that $\pi_j(i)$ represents the interval number of sample i in dimension j .
- (2) For each sample, the sampling points in dimension j can be calculated by Eq. (16):

$$X_{\pi_j(i)}^j = \frac{\pi_j(i) - \mu_{ij}}{n} \quad (16)$$

where μ_{ij} is a random number sampled from a uniformly distributed $U(0, 1)$. By analyzing the equation for generating populations, it can be determined that the LHS method ensures that there must be a sample in each interval by randomly selecting a position in the interval where the sample is located, and can achieve randomness in each sub-interval. The advantage is that it still maintains good performance in high-dimensional spaces, because the sample points in each dimension are generated independently. In order to make up for the shortcomings of

the good point set method in the high-dimensional space, the good point set method and LHS were combined to form a hybrid initialization strategy. The specific process is as follows:

- (1) Generate LHS samples to ensure that the sample points in each dimension are evenly distributed, and enhance randomness by shuffling the order of dimensions.
- (2) Use prime numbers to generate good point set samples, and ensure that the sample points are evenly distributed in the solution space through modular arithmetic.
- (3) Linearly combine the LHS and good point set samples through the dynamic weight α to balance the advantages of the two methods. The calculation of α takes into account the dimension. When the dimension is high, the weight of LHS is relatively increased to make up for the deficiencies of the good point set in high-dimensional spaces. The calculation equation for the dynamic weight α is shown in Eq. (17):

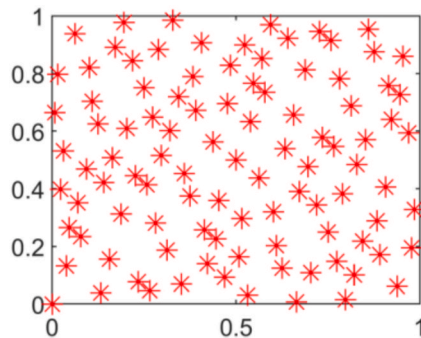
$$X_i^j = \alpha \times P_{L,i}^j + (1 - \alpha) \times P_{G,i}^j \quad (17)$$

$$\alpha = 0.5 + 0.5 \times (1 - 1/\log(m + 1))$$

where X_i^j is the value of the individual i in the dimension j of the population after final fusion. $P_{L,i}^j$ and $P_{G,i}^j$ are the values of the corresponding positions in the initial sample matrix generated by the LHS method and the good point set method, respectively. α is the dynamic weight. This equation ensures that the value range for α is between [0.5, 1]. When the dimension is low, the α is close to 0.5, and the good point set and LHS are given nearly equal weights, making full use of the uniformity of the good point set in the low dimension. With the increase of dimensions, the α monotonically increases, so that the fusion strategy smoothly transitions from “equal emphasis on both” to “LHS-dominated”, so as to take advantage of the stability advantage of LHS in high-dimensional space. This mechanism ensures that a well-distributed initial population can be produced in any dimension.

- (4) Conduct boundary processing on the merged samples to ensure that all sample points are within the defined solution space range.

When the dimension is increased to 100, Fig. 4 intuitively demonstrates the effectiveness of the hybrid initialization strategy. The left-hand and right-hand diagrams in it are the projections of the populations generated by the hybrid strategy and the randomization method respectively in a two-dimensional space. It can be seen that the population generated by the hybrid strategy has better uniformity. Although there is some aggregation in the left-hand diagram, it has an advantage compared to the right-hand diagram. It can also be observed that the projection of the population generated by the hybrid strategy in the two-dimensional space has significantly better uniformity and covers a larger area of the entire space.



Meanwhile, considering that most scholars' improvement strategies for population initialization often remain at adopting different uniform generation strategies, such as chaotic mapping (Yang & Ma, 2023), but there is little research on how to quantitatively measure the uniformity of the initial population. In order to better measure the uniformity of the initial population, this paper introduces the within-population diversity index WD as a measure of overall uniformity (Wang et al., 2024), enabling the comparison of the effectiveness of different strategies through specific numerical values. The calculation equation of WD is as follows:

$$WD(X) = \left[\frac{1}{n} \left(\frac{1}{2} \right)^s - \left(\frac{4}{3} \right)^s + \frac{2}{n^2} \sum_{i=1}^{n-1} \sum_{k=i+1}^n \prod_{j=1}^s \left(\frac{3}{2} - |x_{ij} - x_{kj}| + |x_{ij} - x_{kj}|^2 \right) \right]^{\frac{1}{2}} \quad (18)$$

The smaller the WD value, the better the uniformity of the population. To highlight the effectiveness of the improved strategy, a set of control experiments was designed. The population size was set to 200, and populations were generated through the above two initialization strategies respectively in different dimensions. The experimental results are shown in Fig. 5.

Fig. 5 compares the WD values of the initial population generated by the three initialization strategies in different dimensions. Smaller WD values indicate better uniformity. The results show that the Good Point Set strategy performs well in low-dimensional space, producing the most uniform population. However, when the dimension increases above 10, its performance decreases, even lagging behind the stochastic method. In the high-dimensional space, the proposed hybrid strategy maintains superior and stable performance, validating its necessity for high-

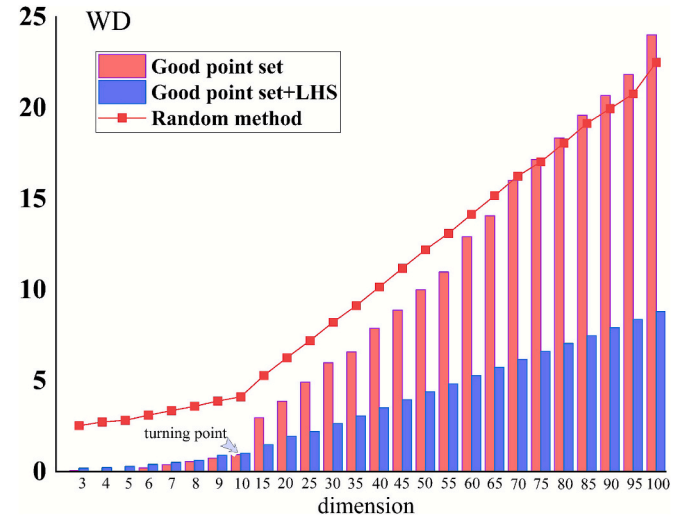


Fig. 5. Comparison of WD values of the three methods.

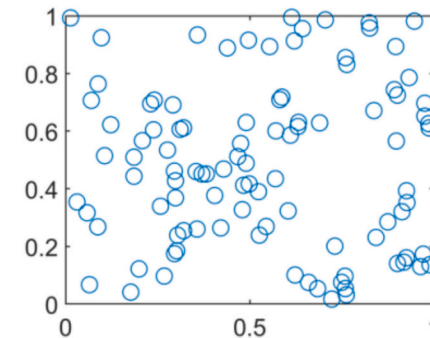


Fig. 4. Comparison of the location distribution of populations (dimension = 100).

dimensional path planning problems.

4.2. Wandering behavior based on population vitality value

In a ship's hold filled with sardines, when several catfish are introduced, the once lackluster sardine shoal can maintain high vitality. This is the "catfish effect". Swarm intelligence algorithms mainly draw on the search mechanisms of biological colonies to find the global optimal value. However, during the search process, once they get trapped in a local optimum, the current optimal value of the population may remain unchanged for several subsequent iterations, and the population becomes "stagnant". The wolf pack algorithm is divided into wandering, summoning, and besieging stages, with global search mainly relying on the wandering stage. Although showing a certain ability to jump out of the local optimal in the process of wandering, the wolf pack algorithm still faces challenges such as stagnation at local optimal solutions and failure to reach the global optimum. Many scholars have noticed this issue and conducted extensive research, attempting to introduce different "catfish" during the search process to enhance the population vitality, such as the Levy flight strategy (Chen, Xi, Wang, & Liu, 2023), adaptive step-size, differential mutation, chaos theory, and random perturbation factors. However, currently, there is no research on how to quantify the population vitality. Therefore, this paper proposes a new concept-the population vitality value H , and presents its calculation equation:

$$H = \frac{t}{3T} + \frac{Y_{front}}{Y_{toulang}} + \frac{Y_{mid}}{Y_{toulang}} + \frac{Y_{back}}{Y_{toulang}} \quad (19)$$

where t is the number of changes of the lead wolf, T is the number of current iterations, Y_{front} is the average fitness of the top 10 % of individuals ranked by fitness of the current population, Y_{back} is the average fitness value of the bottom 10 % of individuals ranked by fitness of the current population, and $Y_{toulang}$ is the fitness value of the current lead wolf.

The proposed population vitality value H is introduced as a composite metric to holistically assess the search state of the population. It transcends existing diversity or stagnation metrics in the evolutionary computation literature by providing a direct link between population state assessment and subsequent search action. Traditional metrics, such as population entropy (Zhang, Pu, & Si, 2021) or the average distance between individuals (Lai, Wu, Peng, Hong, & Cheng, 2025), primarily measure the spatial distribution of the population. While effective in detecting diversity loss, they offer limited insight for prescribing specific remedial search strategies. Crucially, they struggle to distinguish between a population that is healthily converging upon a promising region and one that is prematurely stagnant in a local optimum, as both states can exhibit similarly low spatial diversity. While effective in detecting diversity loss, they may not fully capture the search momentum or the quality gradient within the population.

In contrast, H integrates four informative aspects: (1) the leadership stability, which indicates convergence; (2) the fitness of the elite front Y_{front} , representing the current best-found solution quality; (3) the fitness of the mediocre individuals Y_{mid} , reflecting the overall population level; and (4) the fitness of the worst individuals Y_{back} , signaling the presence of "explorer" individuals. This multi-faceted design allows H to distinguish between a healthily converging population and one that is prematurely stagnant. The practical advantage of H over pure diversity metrics is its role as a decision-making trigger for our adaptive search strategies. For instance, a high value indicates either an ongoing global search with a dynamic leader or a population with a high quality gradient, both of which are desirable states that pure diversity metrics might misinterpret. In this scenario, our algorithm, guided by H , can confidently employ computationally efficient strategies (such as the adaptive step size) to refine the search locally without triggering an unnecessary and costly global exploration phase.

Conversely, a low value, characterized by a stagnant leader and a compressed fitness gradient, unambiguously signals stagnation. This directly triggers the activation of more radical, exploration-focused strategies (such as the Levy flight strategy) to escape the local optimum, even at a higher computational cost. A spatial diversity metric might fail to prompt this necessary response if the population remains spread out in a suboptimal region.

Therefore, H serves as a more nuanced and action-guiding indicator, directly addressing the limitation of using diversity alone for stagnation detection (Blackwell, 2005). It moves beyond mere stagnation detection to enable intelligent stagnation resolution, providing a state-aware mechanism for dynamically allocating computational resources, which is its core practical advantage.

To demonstrate the effectiveness of the H as an indicator of search performance, we first compare the conventional WPA with a single-strategy improved WPA (IWPA) that incorporates only the Levy flight strategy. As shown in Fig. 6. The sub-figure (a) shows the curves of the initial wolf pack algorithm and the improved wolf pack algorithm after 100 iterations, and the sub-figure (b) shows the changing trends of H for the initial wolf pack algorithm and the improved wolf pack algorithm. It can be seen that during the iterative process of the initial wolf pack algorithm, its optimal value has not changed significantly, which means the population may be trapped in a local optimum. Comparing with the right-hand diagram, it can be found that the population vitality shows a significant downward trend. When observing the iterative curve of the improved wolf pack algorithm, in the first 40 iterations, the optimal value changes greatly with an obvious downward trend, indicating that the algorithm is continuously finding better values, and the lead wolf is constantly changing. At the same time, the fitness value of the entire population is also continuously improving. Comparing with sub-figure (b), it is found that in the first 40 iterations, H remains at a relatively high level. After 40 iterations, the algorithm begins to converge, and the population vitality value also starts to decrease.

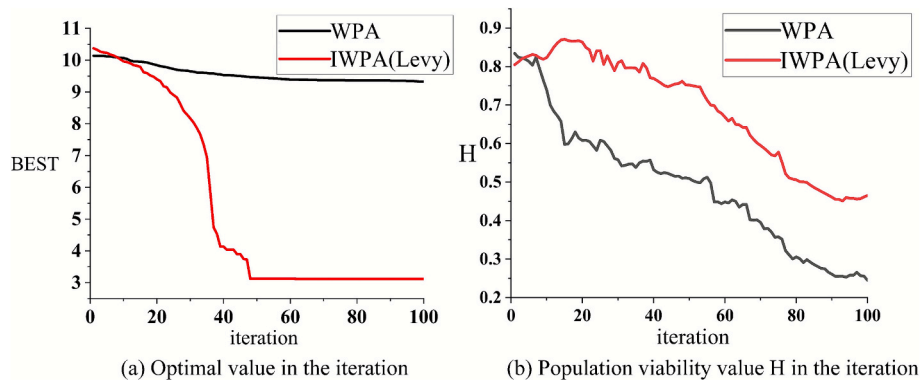


Fig. 6. Comparison of H -values of WPA and IWPA(Levy).

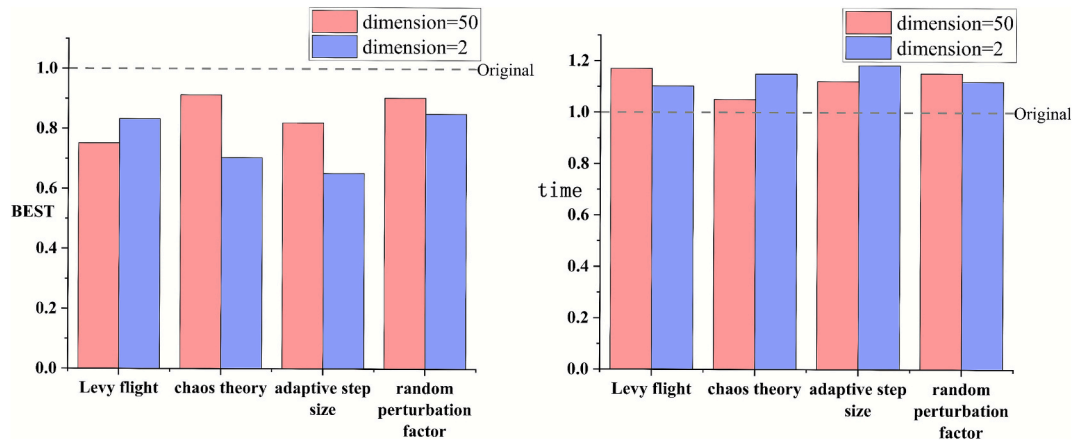


Fig. 7. Performance comparison of the 4 strategies.

The Levy flight strategy, adaptive step size, chaos theory, and random perturbation factor are four common improved search strategies, each having its own advantages and disadvantages in terms of enhancing the algorithm's convergence speed and accuracy. In order to construct the dynamic strategy selection mechanism proposed by us, we must first understand the performance characteristics of each candidate strategy in different computing environments. Through systematic comparative experiments, the trade-offs between convergence accuracy and computational time of four common strategies in low-dimensional and high-dimensional spaces are quantitatively analyzed. This comparative analysis will lay an experimental foundation for subsequent strategy selection rules based on H value. Taking the Sphere function as an example, the optimal value results and running times after 100 iterations for these four strategies in 2-dimensional and 50-dimensional cases are compared. The results of the original algorithm are set as 1, as shown in Fig. 7.

Fig. 7 shows the performance of four candidate strategies on the Sphere function, which is a key prerequisite for formulating dynamic strategy selection rules. The results show that no strategy is optimal in all scenarios, confirming the need for adaptive selection based on the population vitality value. The adaptive step size balances global and local search by dynamically adjusting the step. It performs best in low-dimensional spaces, enabling fast convergence with high precision and strong stability. The logic of step size adjustment is decoupled from the dimension, so the increase in computational load is not obvious in high-dimensional spaces (Chauhan & Rani, 2024; Yang & Ma, 2023). This effectiveness is corroborated by its successful application in novel optimizers (Li, Sun, Yao, & Wang, 2024), such as the RRT-based algorithm for complex search spaces (Lai, Li, & Shi, 2025), and its central role in dedicated UAV path planning algorithms (Wang & Wang, 2026). The Levy flight strategy enhances the global search ability through long-distance migration behavior, but it may cause oscillations due to overly large step sizes. In high-dimensional spaces, Levy step sizes need to be generated frequently, and the step size adjustment logic is complex, leading to a significant increase in computational load (Chen et al., 2023; Yu et al., 2025). Notwithstanding, its power for global exploration is well-established in recent hybrid models, such as its integration into Gravitational Search Algorithms (Joshi, 2023) and Harmony Search for solving complex scheduling problems (Li & Zhou, 2025; Qu, Liu, Zhang, & Li, 2025). The chaos theory avoids premature convergence by increasing population diversity, but excessive chaos may lead to oscillations and insufficient stability. When the dimension is relatively high, the computational cost does not increase significantly (Chen et al., 2023; Wang, 2024). Recent advancements continue to validate this approach; for instance, chaotic mapping has been pivotal in enhancing Mayfly and Dung Beetle optimizers for feature selection (Huang, Yao, Wei, & Zhou, 2024; Sun, Liang, Ding, Xu, & Chang, 2024), and has been directly

applied in an improved sparrow search algorithm for UAV path planning (He & Wang, 2024; Wang et al., 2024a; Wang et al., 2024b), underscoring its direct relevance to our work. The random perturbation factor helps the algorithm escape from local optima by adding random perturbations. If the perturbation intensity is insufficient, premature convergence may occur. In high-dimensional spaces, random perturbations need to be generated independently for each dimension, and the computational load increases linearly with the dimension, yet the calculation for a single perturbation is simple (Dai, Duo, Yuan, & Tang, 2022; Kennedy & Eberhart, 1995). The utility of this strategy is evidenced by its successful implementation in robust path planning solutions, including its use in Particle Swarm Optimization for mobile robots (Tao & Kim, 2024) and in multi-objective optimization for AUV navigation (Sun & Niu, 2024).

Taking the Levy flight strategy as an example, in high-dimensional cases, it can significantly enhance the algorithm's ability to escape from local optima, but the computational cost will also increase substantially. Based on the characteristics of each strategy revealed by the above comparative experiments, we can rationally design the strategy selection rules. When the H value indicates that the population is vigorous but slow to converge, a computationally efficient strategy (such as adaptive step size in low dimensions) can be preferred. When the H -value indicates that the population is low and falls into the local optimum, strategies with strong global exploration capabilities (such as Levy flight strategy) should be used, even if their computational cost is high. Therefore, this paper proposes to design four vitality-value intervals. Different improved search strategies are selected according to the current H value of the population, aiming to balance the convergence speed, search accuracy, and running time, and achieve the optimal result with the minimum cost as much as possible. The schematic diagram of the improved search strategy is shown in Fig. 8.

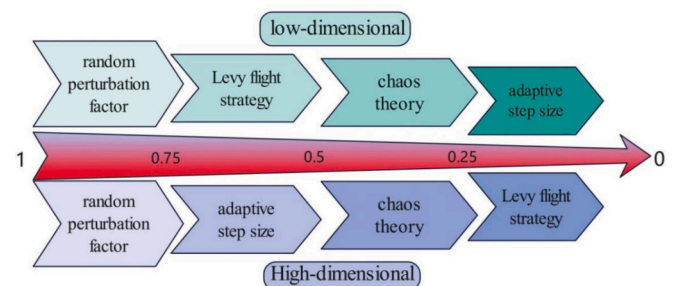


Fig. 8. Schematic diagram of improved wandering behavior.

4.3. Adaptive step size strategy

In nature, when wolves surround and hunt prey, they form a tight encirclement network centered on the prey. The initial wolf pack algorithm adopts a fixed assault step size. This causes wolves close to the prey and those far from the prey to move forward with the same step size simultaneously during the encirclement, leaving a blank area in the middle that gives the prey a chance to escape. Therefore, this paper designs an assault step size based on distance adaptation, and its calculation equation is as follows:

$$step_b^d = rand \times \frac{2(\max_d - \min_d)}{S} \times (c_1 \times \frac{d_i |d_i - d_{pinguin}|}{d_{pinguin}^2} + c_2 \times \frac{d_i}{d_{\max} - d_{\min}}) \quad (20)$$

where $d_{pinguin}$ is the average distance of all wolves from their prey, $d_{\max}$ is the current maximum distance from the prey among the wolves, $d_{\min}$ is the current minimum distance from the prey among the wolves, and c_1 , c_2 represents the weight. The first term of the addition in the brackets reflects the correction of the assault step size by the entire population, while the second term reflects the correction of the assault step size by the wolves at the edge of the population. Through this correction, wolves closer to the prey will obtain a smaller assault step size, while wolves farther from the prey will obtain a larger assault step size. Each individual can choose an appropriate assault step size according to its own situation, which improves the search accuracy and gives the population a greater probability of finding the global optimal solution (see Fig. 9).

The encirclement step size is a crucial parameter in the algorithm, which determines the search accuracy of the wolf pack when besieging the prey (that is, when approaching the optimal solution). In the initial wolf pack algorithm, the encirclement step size is a fixed value, which leads to a lack of flexibility in the search, and there are overlapping search areas among different individuals. To accelerate the convergence of the algorithm, this paper devises an encirclement step size that changes according to the current fitness of an individual, the fitness of the lead wolf, the number of iterations, and the change in the distance from the lead wolf.

$$step_c^d = \frac{(\max_d - \min_d)}{2S} \times \frac{Y_{toulang} - Y_i}{Y_{toulang} - Y_{\min}} \times (1 - \frac{T}{T_{\max}}) \times \frac{d_i}{L_{near}} \quad (21)$$

where $T_{\max}$ is the maximum number of iterations, d_i is the distance of the wolf from the lead wolf, Y_i is the fitness value of the wolf, $Y_{\min}$ is the smallest fitness value in the population, and L_{near} is the determination distance of entering the encirclement. As the number of iterations increases, the closer the individual is to the lead wolf, the smaller the encirclement step size, and the more refined the search will be.

4.4. Population renewal mechanism based on genetic effects

In nature, stronger wolves usually obtain more food and have greater mating rights. This “survival of the fittest” mechanism can ensure that the overall quality of the population is maintained at a high level. Similar ideas are reflected in many swarm intelligence algorithms. For example, in the particle swarm algorithm, when updating the population, the generation of offspring is influenced by the current optimal individual and the positions of the parent generation (Xu, Cao, Du, & Li, 2023). The actual results have also proven that this strategy can accelerate the convergence speed and find the global optimal value. The population update mechanism of the initial wolf pack algorithm adopts the strategy of randomly eliminating R wolves, and the generation of new individuals relies on simple random variables to ensure diversity. It does not fully consider the outstanding contributions of excellent individuals in the population and the genetic influence of parent individuals on offspring individuals. Based on this natural survival law, an improved population update mechanism is designed.

First, taking the fitness of an individual as a reference index, the probability of its generating an offspring individual is calculated. The equation is as follows:

$$p = \left(\frac{n-j}{n}\right)^2 \times [1 + \lg(\frac{Y_i}{Y_{toulang}})] \quad (22)$$

where n is the size of the population, j is the fitness value ranking of the individual in the entire population (from high to low, that is, the lead wolf ranks 1).

Then, R wolves are randomly selected to breed offspring according to the probability. In this way, the mechanism of randomly generating R wolves for elimination and update in the initial wolf pack algorithm is improved. The position of the offspring individual is generated as follows:

$$X = \left(\frac{Y_{toulang}}{Y_{toulang} + Y_i}\right) \times X_{toulang} + \left(\frac{Y_i}{Y_{toulang} + Y_i}\right) \times X_i + rand() \quad (23)$$

where $X_{toulang}$ is the position of the current lead wolf, X_i is the position of the individual. The three added terms respectively reflect the influence of the lead wolf on the offspring individual, the influence of the parent on the offspring individual, and the random perturbation on the offspring individual caused by factors such as the environment. By this method, an offspring population with high initial quality can be generated, which improves the algorithm’s ability to jump out of local optimal solutions, helps it find better path planning solutions, enhances operational efficiency, and ensures the stability and effectiveness of path planning. Fig. 10 vividly demonstrates the improved population update mechanism.

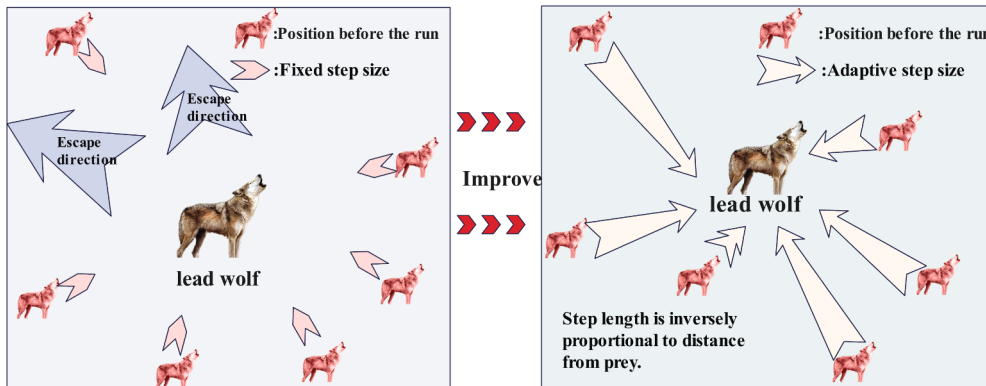


Fig. 9. Schematic diagram of the improved assault steps.

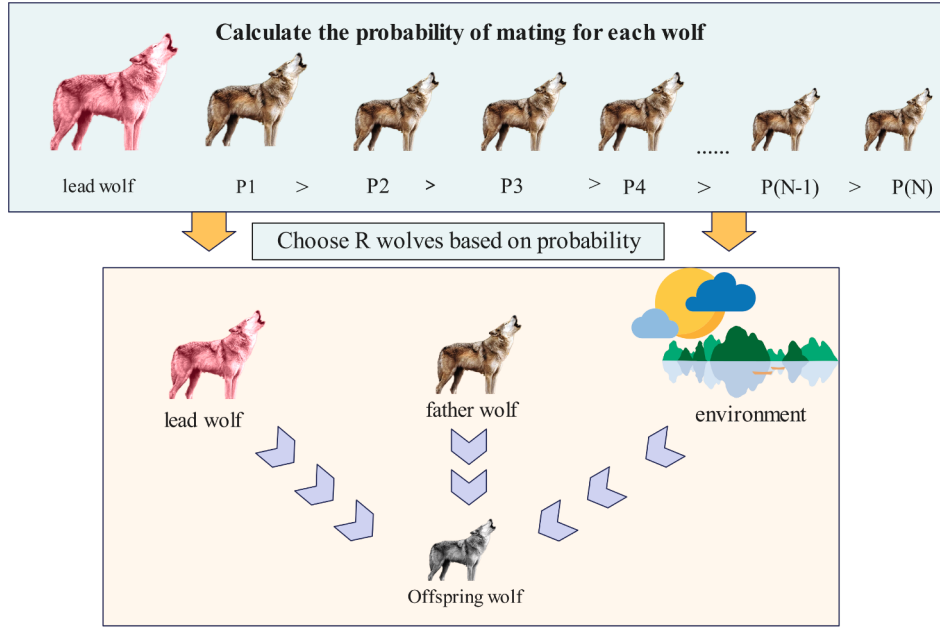


Fig. 10. Schematic diagram of the improved population renewal mechanism.

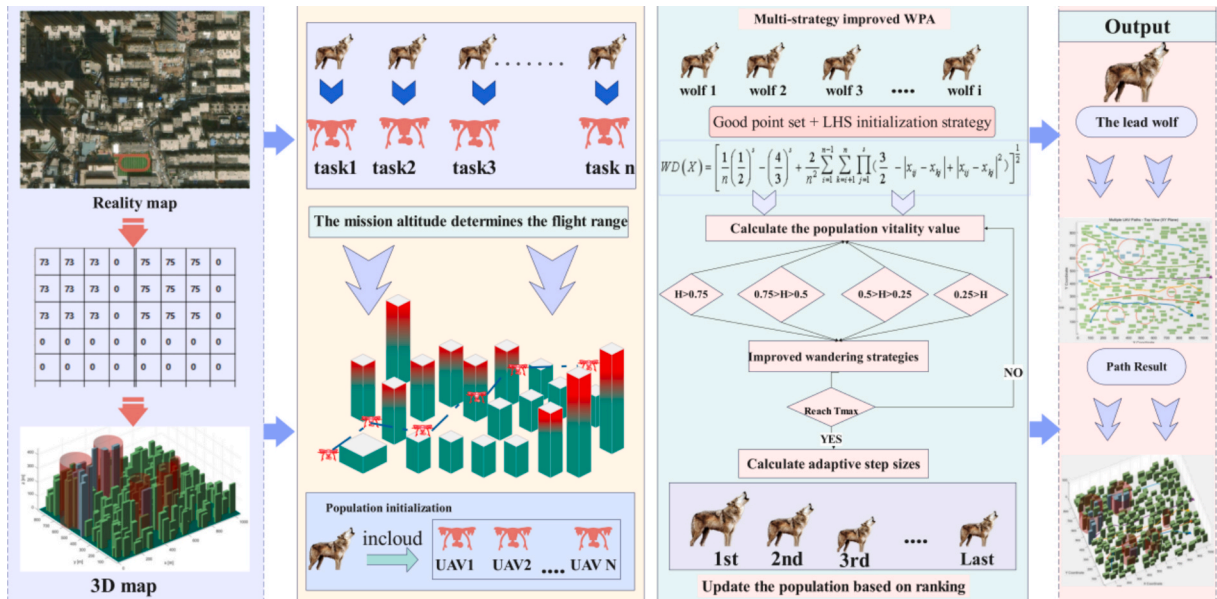


Fig. 11. Structure framework of the text.

4.5. Overall research framework

The overall idea of the MSIWPA proposed in this paper is to enhance the path planning performance of the traditional wolf pack algorithm in complex urban environments by integrating multiple improvement mechanisms. Firstly, through the coordination of multiple strategies, the global search ability and convergence speed are significantly improved. Compared with algorithms such as PSO, its adaptive step size and vitality-value mechanism effectively balance global and local searches. Compared with reinforcement learning methods, it does not require extensive training and has stronger environmental adaptability. Compared with precise planning methods such as MILP, it avoids the combination explosion problem and meets the real-time requirements of online re-planning. These features enable the algorithm to efficiently plan high-quality flight paths that satisfy constraints such as path size,

turning angle, obstacle avoidance, and swarm cooperation in urban areas with dense buildings. The UAV swarm path planning framework based on the MSIWPA is shown in Fig. 11, and the process is as follows:

Step 1: Build the environmental model. Construct a Digital Elevation Model (DEM) based on the geometric parameters of the urban area with dense buildings. Establish a three-dimensional urban space model based on the DEM. Meanwhile, generate a 3D model containing obstacle information and communication interference fields, and mark the starting and ending states of the UAVs.

Step 2: Construct the path cost function. Divide the optimal flight interval based on the altitude requirements of the tasks performed by the UAVs. At the same time, establish an objective function for evaluating the path based on path length cost, path altitude constraints, turning angle constraints, obstacle collision-avoidance

constraints, collision-avoidance constraints among UAVs, and communication maintenance constraints.

Step 3: Solve for the optimal path using the MSIWPA. First, use the population initialization strategy based on the good point set and LHS, where each individual represents the path of the UAV swarm. Then, select the wandering strategy based on the population vitality value. After the wandering ends, accelerate the search speed through the adaptive step size strategy. Finally, update the population based on the genetic effect and output the path of the UAV swarm represented by the lead wolf.

In the framework of solving path planning problems, the internal execution process of the MSIWPA proposed in this paper is the core guarantee for its efficient performance. This algorithm deeply integrates the series of improvement strategies described in Sections 4.1 To 4.4 on the basis of the WPA. The specific calculation process can be systematically explained as follows:

Step 1: Algorithm initialization. Set parameters such as population size n and maximum iteration times $T_{\max}$. Subsequently, the mixed initialization strategy based on the good point set and LHS proposed in Section 4.1 of this article was adopted to generate an initial wolf pack population with uniform distribution and high diversity in the solution space.

Step 2: Fitness calculation and establishment of the lead wolf. Calculate the fitness value of each individual in the population using the cost function constructed in Section 3.3. Select the individual with the best fitness as the lead wolf, and record its path information and fitness value.

Step 3: Wandering behavior. Select several wolves with better fitness as scout wolves, except for the lead wolf. Each scout wolf explores in h directions around it and dynamically selects a wandering strategy based on the population vitality value proposed in Section 4.2. If the wolf finds a position better than the current lead wolf, they will become the lead wolf and perform summoning actions; Otherwise, continue wandering until reaching the maximum number of wandering before turning into an assault wolf.

Step 4: Summoning behavior. The lead wolf summoned the pack of wolves, and the surrounding assault wolves quickly approached the position of the lead wolf with an assault step. During this process, the assault step size adopts the adaptive assault step size designed in Section 4.3, which dynamically adjusts according to the distance between the individual and the lead wolf to achieve fast and accurate encirclement.

Step 5: Besieging behavior. The wolf pack launches a final encirclement on the prey represented by the position of the lead wolf. The encirclement step size of each wolf adopts the adaptive encirclement step size proposed in Section 4.3, which decreases with the increase of iteration times and is related to the distance and fitness between

the individual and the lead wolf, thus achieving more refined local search in the later stage of iteration.

Step 6: Population renewal. According to the principle of “survival of the fittest”, eliminate the R wolves with the worst fitness in the population. Subsequently, based on the genetic effects based population updating mechanism proposed in section 4.4, individuals are assigned corresponding reproductive probabilities according to their fitness rankings, and new offspring individuals are generated through Eqs. (22) and (23) to maintain population diversity and vitality.

Step 7: Termination condition judgment. Check whether the algorithm has reached the maximum number of iterations or meets the convergence accuracy requirements. If satisfied, output the optimal path solution represented by the current lead wolf; Otherwise, increase the number of iterations by one and return to Step 2 to continue iterating.

The above process together constitutes the core engine of the MSIWPA. Through the synergistic effect of population initialization, wandering behavior based on population vitality value, adaptive step size, and population renewal mechanism based on genetic effects, this algorithm demonstrates excellent global exploration ability, fast convergence speed, and excellent local development accuracy in complex path planning solution spaces. The overall execution process of the algorithm is shown in Fig. 12.

The pseudo-code of MSIWPA is given below in Algorithm 1.

5. Experiments and analysis

5.1. UAV swarm path planning based on MSIWPA

UAV path planning possesses the features of nonlinearity, discretization and high dimensionality. In this paper, the MSIWPA algorithm is designed to provide solution for UAV swarm path planning in the urban environment with dense buildings in a 3D map. By analyzing the paths generated by MSIWPA, its capability to solve the path planning problem is presented.

In our preliminary studies, we observed that the conventional WPA often fails to converge to feasible solutions in the high-dimensional and heavily constrained UAV path planning problem considered here, due to premature convergence and insufficient exploration. It is noteworthy that the proposed MSIWPA belongs to the class of model-free, training-free meta-heuristic algorithms. While deep learning methods (e.g., DRL, GNNs) have shown remarkable success, they often necessitate substantial offline training and may face challenges in generalization and real-time adaptation in highly dynamic and uncertain environments. The MSIWPA, with its adaptive step-size and population vitality mechanisms, is designed specifically for such online planning scenarios, providing a robust and immediately deployable alternative without the need for any prior training phase. Therefore, the conventional WPA is

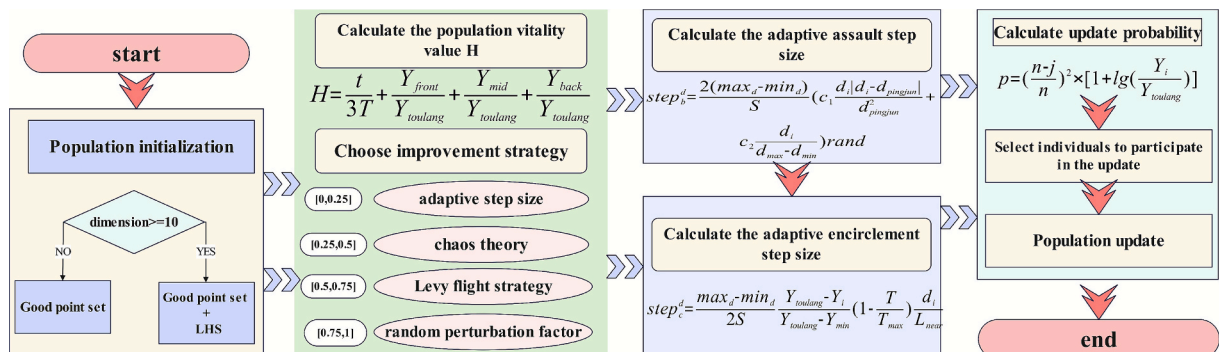


Fig. 12. Algorithm flowchart.

Table 2

The start and end of the 6 UAVs.

Coordinate	Start	Target
UAV1	(100, 100, 65)	(900, 150, 120)
UAV2	(100, 200, 100)	(950, 250, 90)
UAV3	(90, 420, 60)	(960, 350, 120)
UAV4	(90, 450, 65)	(1050, 450, 50)
UAV5	(50, 790, 55)	(950, 578, 90)
UAV6	(150, 850, 55)	(150, 850, 55)

not included in the main comparative analysis. Instead, we focus on comparing the proposed MSIWPA—which integrates multiple enhancement mechanisms—against other state-of-the-art swarm intelligence and machine learning algorithms to demonstrate its competitive advantage in complex urban environments. In this paper, the MSIWPA algorithm is compared with five other algorithms. The selection of these baseline algorithms is deliberate and comprehensive, designed to ensure a fair and rigorous benchmarking. They represent a panel of state-of-the-art meta-heuristics published in recent, high-impact literature. This panel encompasses a diverse range of inspirations, including bio-inspired algorithms (COA (Dehghani, Montazeri, Trojovská, & Trojovský, 2023), SFOA (Zhong et al., 2025), a physics-inspired algorithm (TOC (Braik et al., 2025), and a machine-learning-based algorithm (IBL (Mirrashid & Naderpour, 2023). Furthermore, CFOA (Jia, Zhou, Zhang, & Mirjalili, 2025) was included as a highly competitive bio-inspired optimizer. By benchmarking against this suite of contemporary and varied algorithms, we aim to robustly demonstrate the competitive advantage and general efficacy of our proposed MSIWPA in solving the complex UAV swarm path planning problem. All algorithms are executed 30 times, with a maximum of 100 iterations. The population size is set to 30, and the remaining parameter settings are based on the reference values in the literatures where the algorithms were proposed. The obtained paths and the final path costs are used to compare their performance.

In the design scenario, we consider a path planning problem with 6 UAVs. Each individual in the population contains the path of 6 UAVs, and the number of distance assessment points is consistent with the number of waypoints. In addition, both $\theta_{turnmax}$ and $\theta_{climbmax}$ are stipulated to be 45° , the value of D_R is 50 m, the communication impact factor γ is 0.3, r_{safe} is 10 m, and the size radius of the UAV is designed to be 0.3 m. The relevant parameter settings for calculating the cost of communication constraints are: $c = 3 \times 10^8$ m/s, $f_c = 2.4$ GHz, $p_t = 26$ dBm, $p_n = -110$ dBm, $\omega = 0.1$, $\eta = 0.22$, $SNR_t = 2.5$. The starting and ending points of each UAV are shown in Table 2.

The selection of key parameters in our experiments was guided by domain-specific standards, preliminary pilot studies, and empirical tuning to ensure realism and algorithmic robustness.

$r_{safe} = 10$ m: This value accounts for the physical radius of the UAV (0.3 m) and incorporates a conservative buffer to accommodate positioning errors, sensor noise, and sudden maneuvers. It aligns with typical UAV safety standards and urban operation guidelines, which often recommend a minimum separation of 5–15 m for small UAVs in dense environments.

$D_R = 50$ m: This value mainly refers to the measured experience and relevant literature on the interference of UAV communication frequency bands in urban environment (Stahlbuhk et al., 2022). Specifically, measured studies have shown that in the common frequency bands of UAVs such as 2.4 GHz and 5 GHz, the communication interference

Table 4

Parameters for setting the communication interference field.

Field	S1	S2	S3	S4	S5
center	(100, 600)	(300, 150)	(415, 650)	(740, 330)	(530, 150)
radius	110	70	60	35	70
height	180	180	150	120	150

intensity in the 50-meter low-altitude area can be significantly enhanced compared with the ground. This setting is designed to simulate typical strong interference scenarios that drones may encounter in the flight altitude range in dense urban electromagnetic environments. At the same time, this value is also consistent with the interference influence range considered in the study of UAV swarm communication to deal with malicious interference and ensure the information rate of the backhaul link.

$\theta_{turnmax}$ and $\theta_{climbmax} = 45^\circ$: the setting is primarily based on the physical maneuvering limitations of multi-rotor UAVs, task requirements, and widely adopted empirical values in the industry (Shen, Zhang, Li, & Li, 2025).

To further verify the robustness of these parameter selections, local sensitivity analysis was performed. Key parameters, including r_{safe} and D_R , were disturbed by 10 % compared to the nominal value. The results show that the performance change of the algorithm measured by runtime is less than 3.5 % under the experimental settings in Table 2, confirming that the proposed MSIWPA is not extremely sensitive to subtle changes in these parameters and shows strong stability. As shown in Table 3.

To simulate the real-world urban environment with dense buildings, the experiment takes a residential community southwest of Beidajie Subway Station in Xi'an, Shaanxi Province as the experimental simulation environment. To simplify the environmental modeling process, only medium-and large-sized buildings are selected to construct the model, and gardens, guard booths, etc. are not modeled. At the same time, to simulate the electromagnetic interference present in the urban environment, five communication interference fields are set in the model, and their relevant parameters are shown in Table 4.

The model is distributed within a terrain area of 1045×879 . Also, for intuitive presentation, the actual altitude data is amplified proportionally, which does not affect the experimental results, as shown in Fig. 13. In the figure, the light blue parts represent buildings with an actual height greater than 40 m, and the green parts represent buildings with an actual height less than 40 m.

It is important to note that although the current experiments are conducted in a static environment, the design features of the proposed MSIWPA give it great potential for dynamic path planning. The key to handling dynamic obstacles lies in the algorithm's ability to perform efficient online replanning. The wandering behavior based on population vitality value enables fine motion adjustments when encountering new obstacles. More importantly, the innovative proposed population vitality value H reflects population diversity and search stagnation. When the H -value drops suddenly, a globally reinforced wandering phase can be triggered, effectively enabling the swarm to "rediscover" the environment when the original path is blocked.

In dynamic scenarios, the environmental model will be updated in real-time via onboard sensors or ground controls. MSIWPA can then be called to perform fast path re-optimization from the current position of the UAV. The high convergence rate and resolution quality

Table 3

Parameter sensitivity analysis.

Parameter	A	B	C	D	E	F	G	H	I
r_{safe}	9	9	9	10	10	10	11	11	11
D_R	45	50	55	45	50	55	45	50	55
Runtime (s)	41.9	42.1	41.8	43.6	42.5	43.7	44.1	43.8	43.5

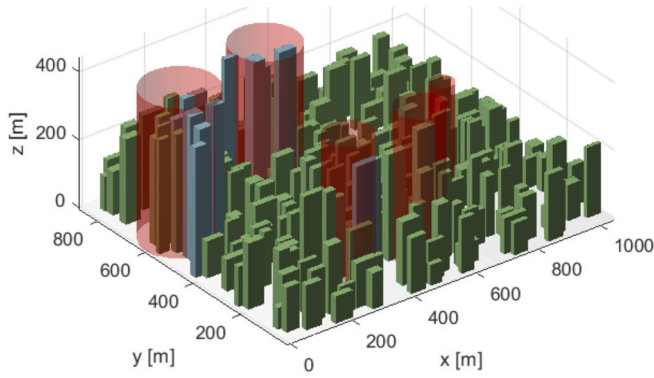


Fig. 13. Urban building model.

demonstrated by us in static experiments suggests that MSIWPA is a promising candidate for such online reprogramming tasks. Future work will include integrating dynamic obstacle prediction models and testing algorithm performance in real-time simulation platforms.

5.2. Analysis of path planning results

The comparison of the costs of each UAV in the final results of the six algorithms is shown in Fig. 14. Considering the randomness of the search processes of each algorithm, the paths corresponding to the experiment whose final cost function value ranks 15-th among the 30 experiments are selected for plotting. It can be seen that, compared with other algorithms, the costs of each UAV in the final results obtained by the MSIWPA algorithm are all the minimum, which reflects that the paths obtained by it are superior to those of other algorithms.

In the objective function, the cost of maintaining distance within the swarm and the cost of communication constraints only take two values, 0 and infinity. Since all algorithms find feasible solutions in the final results, the values of both are 0. Therefore, the sum of the costs of each UAV in Fig. 14 is the overall cost of the UAV swarm path. By analyzing the differences in the costs of each UAV in the final results of the six algorithms, it can be concluded that the value of the overall objective function of the final path results of the MSIWPA algorithm must also be the smallest. Especially compared with the COA algorithm, it shows significantly stronger global search ability. Fig. 15 vividly demonstrates the path results of the MSIWPA algorithm.

Through the display of the flight path, the superior performance of MSIWPA is strongly demonstrated. It can be observed that there are no

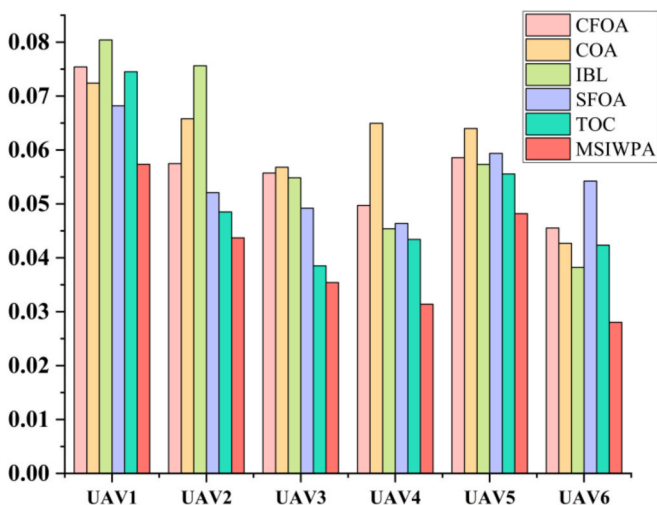


Fig. 14. Comparison of UAV costs of each algorithm result.

sharp turning angles in the paths of the six UAVs. Meanwhile, the paths generated by MSIWPA are compact without collisions, which better ensures the communication connections among the UAVs, and the path lengths are relatively shorter. From the top-view, some of the UAVs have flown over some buildings at a height above the buildings. However, when combined with the 3D view, it can be found that parts of the paths of each UAV pass through the interior of the buildings. This is because when evaluating the obstacle-avoidance cost, only the waypoints are evaluated, and it is still possible for the path segments to pass through obstacles. Therefore, the value of the overall objective function is not infinite.

But a visual comparison of paths in different local environments is insufficient to claim the superior smoothness of one algorithm over another. To address this with rigor, we introduce two quantitative metrics for an objective assessment of path smoothness.

Average Turning Angle. The mean of the absolute turning angles. A smaller value indicates smoother horizontal maneuvers and less aggressive heading changes.

Average Climbing Angle Change. The mean of the absolute difference in climbing angles between consecutive path segments. A smaller value indicates smoother vertical transitions and greater stability in altitude control.

These metrics were computed for the final paths generated by each algorithm in the 30 independent runs. The statistical results are presented in Table 5.

The data in Table 5 unequivocally demonstrates that MSIWPA outperforms all other algorithms in both smoothness metrics. The paths planned by MSIWPA exhibit significantly smaller turning angles and more gradual changes in climb angle. This is a direct consequence of the algorithm's effective integration of the turning angle cost and the altitude cost within the objective function, coupled with the refined search capabilities of the adaptive step size and population renewal strategies. Therefore, we can objectively conclude that MSIWPA possesses a stronger optimization capability for generating smoother and more flyable paths, rather than merely finding a path that traverses a naturally smoother corridor.

In Fig. 16, the left figure compares the average flight altitude of the six UAVs in the results of each algorithm. It can be seen that the altitude change of the MSIWPA algorithm is the smallest, and the flight altitudes of the six UAVs are relatively stable. This reflects that the MSIWPA algorithm can plan the paths of each UAV within the optimal flight altitude range, meaning it has a higher solution accuracy. The paths planned by other algorithms also show a concentrated trend in flight altitude, all relatively concentrated around 70 m. This is because the layering strategy divides the flight space into two layers, and flying at too high or too low an altitude will result in a high altitude cost. Therefore, the flight altitudes of the results of each algorithm strive to approach the optimal flight altitude range. The right figure compares the maximum altitude differences of the paths of each UAV in the results of each algorithm. It can be seen that the relevant value of the MSIWPA algorithm is the smallest, indicating that the flight altitude change of the UAVs is the smallest, so its altitude cost is also lower.

In order to quantitatively evaluate the performance of different algorithms in UAV swarm path planning, this study conducted an in-depth statistical analysis on the flight altitude data of six UAVs. As shown in Table 6, we calculate the key indicators such as the average flight altitude, standard deviation (SD), coefficient of variation (CV) and maximum altitude difference within the swarm under each algorithm.

The statistical analysis results clearly show that the MSIWPA algorithm proposed in this study has significant advantages in maintaining the flight altitude stability of the swarm. Specifically, the altitude standard deviation and coefficient of variation of the MSIWPA planning path are the lowest among all comparison algorithms, which statistically proves that the algorithm can make the altitude distribution of UAVs in the swarm the most concentrated, so as to achieve the best cooperative flight stability. To test the statistical significance of these differences, we

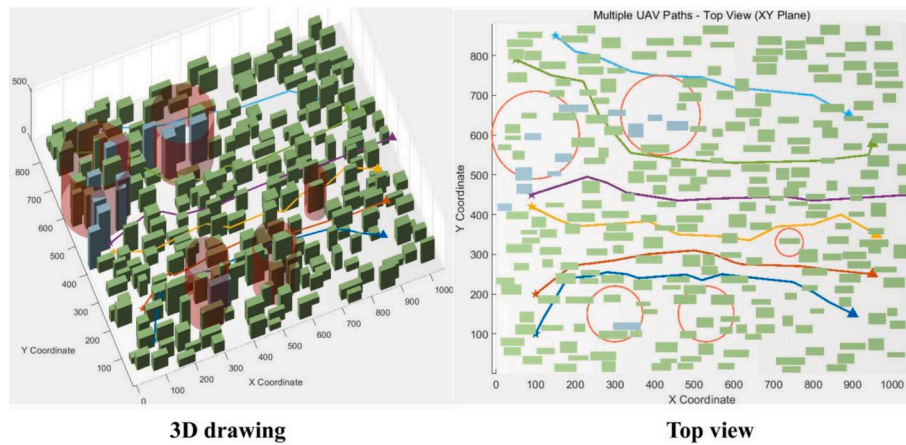


Fig. 15. Path results.

Table 5
Quantitative comparison of path smoothness metrics.

Algorithm	Average Turning Angle (°)	Average Climbing Angle Change (°)
MSIWPA	12.7	5.3
COA	28.4	11.2
CFOA	24.1	9.8
SFOA	31.5	13.5
IBL	35.2	15.1
TOC	26.8	10.6

performed a one-way analysis of variance, and the results ($F = 4.82$, $p = 0.0032$) showed that the influence of different algorithms on the high stability of the swarm was highly statistically significant. The subsequent Tukey HSD post-hoc test further confirmed that there was a significant difference between MSIWPA and each comparison algorithm, as all values of p were less than 0.05, providing a solid statistical inference basis for the superiority of MSIWPA.

In addition, this study focuses on the amplitude of altitude variation between different UAVs, rather than the altitude fluctuations of the same UAV over time. This methodological choice stems from the core requirement of swarm collaborative path planning: in dense urban environments, maintaining a relatively high degree of consistency between UAVs within the swarm is crucial to ensure stable communication links (especially under NLoS conditions) and avoid collisions within the swarm. Although the instantaneous smoothness of the flight altitude of individual UAVs affects its energy consumption and controllability, it cannot effectively characterize the synergistic efficiency and overall safety of the entire swarm. MSIWPA's outstanding performance in this

Table 6
Statistical analysis of flight altitude stability of six algorithms.

Algorithm	Mean Altitude (m)	Std. Deviation (m)	Coefficient of Variation (%)	p-value (vs. MSIWPA)	Max Altitude Difference (m)
MSIWPA	76.17 ± 5.15	5.15	6.76	—	18
COA	73.83 ± 8.72	8.72	11.81	0.023	24
CFOA	77.00 ± 12.84	12.84	16.68	0.045	37
SFOA	75.67 ± 12.35	12.35	16.32	0.038	35
IBL	73.00 ± 14.02	14.02	19.21	0.015	39
TOC	75.00 ± 11.36	11.36	15.15	0.028	30

Table 7
Runtime Performance Comparison.

Algorithm	Average running time(s)	rank
MSIWPA	42.5	3
COA	58.2	5
CFOA	38.5	2
SFOA	32.1	1
IBL	135.6	6
TOC	51.3	4

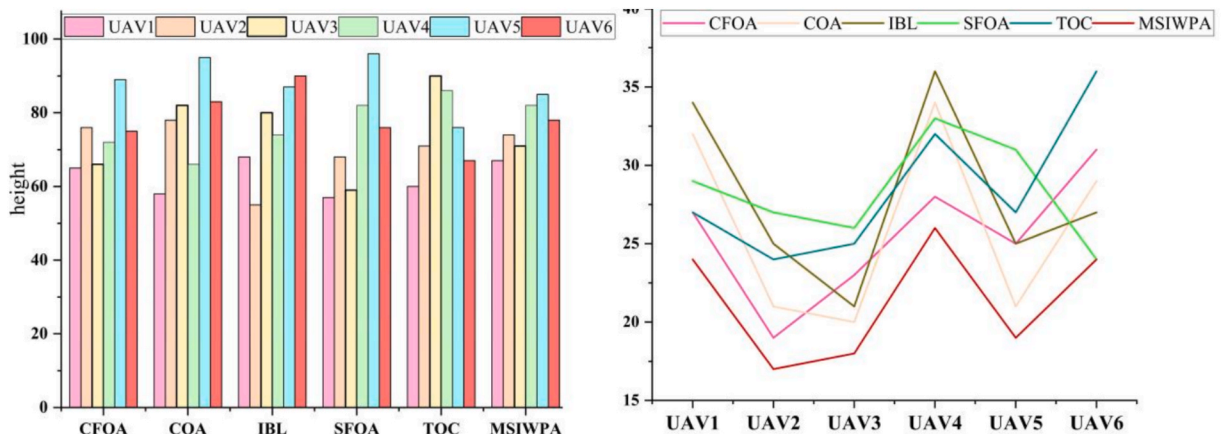


Fig. 16. High-level analysis of the results of each algorithm.

indicator—its maximum height difference within the swarm is only 18 m, which is far lower than that of other algorithms, which directly reflects its excellent swarm-level collaborative planning ability under complex constraints.

This advantage can be attributed to the multi-policy improvement mechanism employed by MSIWPA. The wandering behavior based on population vitality value enables the algorithm to adaptively switch between exploration and development, avoiding precocious population maturity, so as to find a more globally optimized solution in the three-dimensional space. The population renewal mechanism based on genetic effects effectively retains high-quality individuals and ensures the overall evolutionary direction of the population. These mechanisms work together to enable MSIWPA to plan flight trajectories that not only have the lowest total path cost, but also have the best swarm synergy under the premise of meeting various constraints.

For practical deployment of UAV swarm, computational efficiency is as crucial as solution quality, particularly for mission pre-planning and potential real-time applications. To comprehensively evaluate this aspect, we conducted a rigorous runtime analysis across all six algorithms in the 6-UAV path planning scenario. The quantitative results are systematically presented in Table 7, which reveals distinct computational characteristics among the evaluated algorithms. MSIWPA achieves a balanced runtime performance of 42.5 s, ranking third among the six algorithms. This positioning reflects the computational overhead introduced by its multi-strategy mechanisms—including population vitality assessment, adaptive step sizing, and genetic-based population renewal—which are precisely the features responsible for its superior path planning quality demonstrated in previous sections.

Several key observations emerge from this analysis. The IBL exhibits significantly higher computational demands (135.6 s), being approximately 3.2 times slower than MSIWPA, which limits its practicality for time-sensitive applications. Conversely, while SFOA (32.1 s) and CFOA (38.5 s) demonstrate faster execution—ranking first and second respectively—this computational advantage comes at the substantial cost of inferior path quality. MSIWPA strikes an optimal balance, delivering significantly better solution quality than the faster algorithms while maintaining reasonable computational requirements that are 27.1 % faster than COA and 17.2 % faster than TOC. The 42.5 s runtime of MSIWPA falls well within acceptable limits for off-line mission planning applications. For scenarios requiring faster response times, several optimization pathways exist: parallelization of independent operations, implementation of early termination criteria based on convergence detection, and utilization of the hierarchical framework discussed in Section 6. This analysis confirms that MSIWPA provides an excellent trade-off between computational efficiency and solution quality, making it a practical choice for real-world UAV swarm path planning applications.

To better evaluate the performance of MSIWPA, the number of communication interference fields is increased to 8. The relevant settings of the additional 3 communication interference fields are shown in Table 8, and the experimental results are presented in Fig. 17 and Fig. 18.

Even when the number of communication interference fields is increased, the MSIWPA algorithm can still obtain better path results. As can be seen from Fig. 17, the MSIWPA algorithm can find paths with lower costs in the path planning of each UAV, indicating its higher search accuracy. Analyzing the top-view of the final path results of the MSIWPA algorithm, compared with the path results in Scenario 1, even

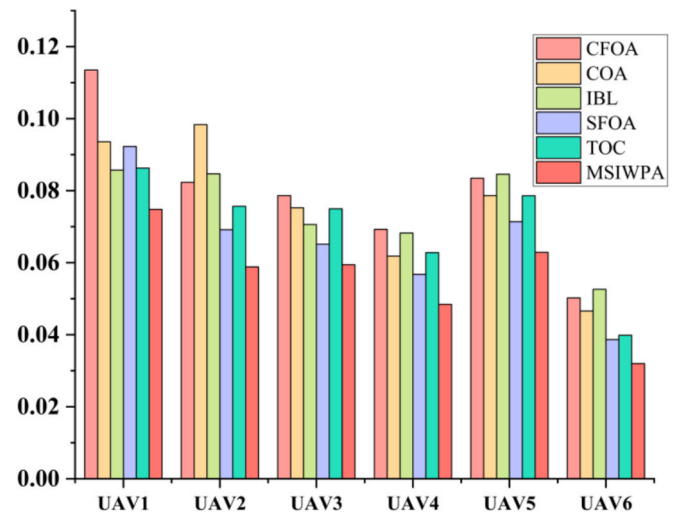


Fig. 17. Comparison of UAV costs in the results of each algorithm.

after increasing the number of communication interference fields, each UAV can still plan a path while maintaining communication. As can be seen from the 3D view, some of the path segments that appear to pass through buildings in the top-view actually do not, but pass over the buildings. By incurring a small altitude cost for some parts of the path, MSIWPA avoids frequent turning between buildings, so the cost of the entire path is actually reduced. Similarly, we used two quantitative metrics to objectively assess path smoothness, and the statistical results are shown in Table 9.

The data in Table 9 clearly shows that MSIWPA still outperforms all other algorithms on these two smoothness metrics even with an increase in the number of communication interference fields. The MSIWPA-planned path exhibits significantly smaller turning angles and slower climb angle changes. Even in the face of more complex environments, MSIWPA can still find smoother and more flyable paths, proving its powerful optimization capabilities.

In Fig. 19, by comparing the average flight altitude and the maximum altitude difference of each UAV, it shows that after increasing the communication interference fields in the environment, the final path obtained by the MSIWPA algorithm remains the most stable. Even though the maximum altitude difference of UAV5 is not the smallest, its overall performance still ranks the first.

In order to quantitatively evaluate the performance of different algorithms in the path planning of UAV swarms, the flight altitude data of 6 UAVs were also analyzed in depth under the condition of increasing the communication jamming field, as shown in Table 10.

After increasing the number of communication interference fields, the MSIWPA algorithm still shows significant superiority. Specifically, the altitude standard deviation and coefficient of variation of the MSIWPA planning path are the lowest among all comparison algorithms, which indicates that even in the environment of high interference, MSIWPA can still maintain the most concentrated altitude distribution of the UAV swarm, thereby maintaining excellent cooperative flight stability.

After increasing the number of communication interference fields, the algorithm runtime is evaluated, and as shown in Table 11, the average runtime of all algorithms increases as expected compared to the environment without the interference field. However, MSIWPA is still very competitive in terms of time efficiency, and its runtime is at the shortest level among all comparison algorithms.

This result strongly echoes and unifies the stability advantage mentioned above: MSIWPA does not sacrifice computational time in exchange for the quality of the solution, but outputs a path planning scheme with better synergy and stability in a shorter or acceptable time.

Table 8

Setting parameters of communication interference fields.

Field	S6	S7	S8
Center	(600, 300)	(750, 200)	(720, 720)
radius	30	40	50
height	150	150	150

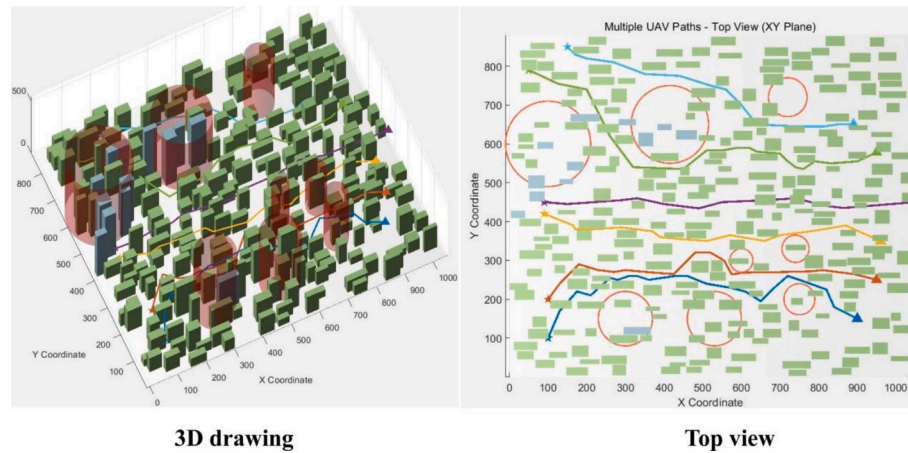


Fig. 18. Path results.

Table 9

Quantitative comparison of path smoothness metrics.

Algorithm	Average Turning Angle (°)	Average Climbing Angle Change (°)
MSIWPA	13.4	6.1
COA	29.5	13.6
CFOA	26.9	11.5
SFOA	30.8	10.7
IBL	36.7	13.7
TOC	28.3	11.8

This excellent comprehensive performance is attributed to the intrinsic efficiency of its multi-strategy improvement mechanism. The wandering behavior based on the population vitality value enables the algorithm to intelligently balance “exploration” and “development”, avoiding lengthy blind searches in invalid regions, so as to faster approach the global optimal region in the complex three-dimensional solution space. At the same time, the population renewal mechanism based on genetic effects accelerates the convergence process of populations by retaining and utilizing high-quality genes.

Therefore, even in highly complex environments with communication jamming fields, MSIWPA demonstrates excellent time-performance trade-offs. There is no surge in running time and poor solution quality due to local optimization; and no synergistic stability of the scheme is sacrificed in pursuit of ultimate speed. From the perspective of computational efficiency, it further consolidates the comprehensive superiority and practical application value of MSIWPA in dealing with UAV swarm path planning problems in complex urban environments.

5.3. Discussion on objective function design and trade-offs

The design of the comprehensive objective function in Eq. (12) is pivotal to the success of our path planning model. A relevant discussion on its component necessity and inherent trade-offs is warranted. It is important to note that several constraints, namely obstacle avoidance, intra-swarm collision avoidance, and communication maintenance, are implemented as hard constraints through infinite penalty terms. This design choice inherently prioritizes safety and feasibility above all else, representing the most critical trade-off. The empirical results strongly validate this design. For instance, the successful paths in Figs. 15 and 18 inherently demonstrate a balance: they are not only short and smooth but also strictly avoid collisions and maintain communication links, as evidenced by the zero cost for C_j^S and C_j^I in the final feasible solutions. This outcome would not be achievable if any of these hard constraints were removed, effectively serving as an ablation study. For instance, without the communication constraint, a shorter path through an interference field might be selected, but it would be infeasible due to packet loss. Similarly, the altitude cost C_{ij}^H acts as a soft constraint that guides the planner towards the energy-efficient flight layer. The concentration of flight altitudes around 70 m in Fig. 16 demonstrates the algorithm's effective balancing of path length and altitude-related energy consumption. A formal sensitivity analysis of the weighting factors for these soft constraints represents a valuable direction for future work to further refine the solution Pareto front.

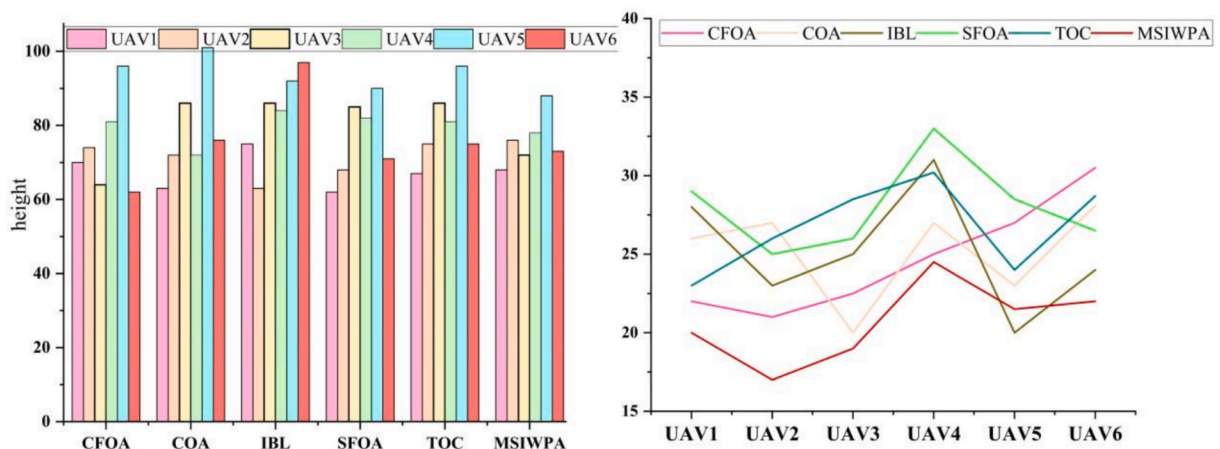


Fig. 19. Altitude analysis of each algorithm result.

Table 10
Statistical analysis of flight altitude stability of six algorithms.

Algorithm	Mean Altitude (m)	Std. Deviation (m)	Coefficient of Variation (%)	p-value (vs. MSIWPA)	Max Altitude Difference (m)
MSIWPA	76.98 ± 5.83	5.83	6.93	—	20
COA	74.65 ± 9.12	9.12	12.14	0.026	28
CFOA	78.15 ± 11.57	11.57	17.58	0.042	32
SFOA	73.86 ± 12.87	12.87	17.26	0.035	37
IBL	74.73 ± 13.17	13.17	18.27	0.018	42
TOC	75.61 ± 12.67	12.67	16.37	0.027	34

Table 11
Runtime Performance Comparison.

Algorithm	Average running time(s)	rank
MSIWPA	56.4	3
COA	72.4	5
CFOA	46.2	2
SFOA	38.7	1
IBL	174.5	6
TOC	60.7	4

6. Summary and outlook

This paper proposes a multi-strategy improved wolf pack algorithm and successfully applies it to the path planning of UAV swarms in urban environments with dense buildings. An urban electromagnetic scenario with dense communication interference fields is constructed, and an overall objective function is designed by combining various path-constraint conditions. An initialization strategy that combines the good point set and LHS is introduced to improve the quality of the initial population. The improved wandering strategy based on the population vitality value can adaptively select an improved search method, better balancing the requirements of computational complexity and computational accuracy. The improved adaptive assault step size and encirclement step size can effectively accelerate the convergence speed of the algorithm. A population-update strategy based on the genetic mechanism is designed to retain high-quality individuals and keep the population quality at a high level. Simulations and comparisons of UAV swarm path planning are carried out in specific scenarios of urban environments with dense buildings. The results show that MSIWPA can more effectively plan high-quality flyable paths for UAV swarms.

Looking beyond the immediate technical contributions, this research offers a significant step toward bridging the gap between controlled optimization and real-world autonomy. While the current work provides a robust solution for static environment planning, it also lays a foundational framework for confronting the more formidable challenge of dynamic urban airspace navigation. The inherent adaptability of MSIWPA's design—its responsiveness to current population state through the vitality value and its fine-grained local search via adaptive steps—suggests a strong potential for extension to dynamic settings. Building upon this foundation, our future research will be directed towards addressing the following fundamental, interconnected challenges.

- (1) From Static to Dynamic: Theoretical Foundations for Online Replanning. A core limitation of our current model is its confinement to static environments, which cannot model the stochastic and non-stationary nature of real urban airspace. Our future work will therefore focus on formulating the path planning

problem as a sequential decision-making process under uncertainty. This entails the integration of real-time sensor data streams with predictive models for dynamic obstacle (e.g., other UAVs, manned aircraft) trajectory forecasting. The key theoretical challenge is to evolve MSIWPA from a single-shot optimizer into an online, incremental replanning algorithm. We will investigate mechanisms for warm-starting the optimization from the previous solution and dynamically adjusting the search space based on predicted changes, thereby achieving a tractable computational footprint for real-time operation. This direction inherently bridges the fields of meta-heuristic optimization and stochastic optimal control.

- (2) Enhanced Environmental Fidelity: Integrating Multi-Physics Models. Our current environmental model relies on simplistic geometric and empirical approximations, a significant simplification that must be transcended to transition from simulation to reality. To address this, we plan to incorporate high-fidelity, multi-physics urban digital twins. This includes integrating computational fluid dynamics models for urban wind fields, which critically impact UAV energy consumption and stability, and ray-tracing-based wireless channel models for precise prediction of non-line-of-sight conditions and packet loss. By doing so, the objective function will evolve to optimize not just for geometric feasibility but for energy resilience and communication robustness in a physics-grounded manner.
- (3) Algorithmic Scalability and Hybrid Intelligence. The centralized optimization approach adopted here presents a scalability bottleneck when extended to larger swarms and more complex environments. A pivotal future direction is therefore the investigation of hybrid intelligence architectures. This explores the synergy between the global optimization strength of MSIWPA and the perceptual and decision-making capabilities of Deep Reinforcement Learning. One promising avenue is a hierarchical framework, where a high-level DRL policy makes macro-decisions, and a low-level MSIWPA module performs fine-grained, constraint-satisfying path optimization within those corridors. Furthermore, we will explore decentralized versions of MSIWPA to enhance the system's scalability and robustness.

By pursuing these directions, we aim to advance the state-of-the-art from solving a well-defined optimization problem towards creating autonomous UAV swarms capable of long-term, safe, and efficient operation in the open-ended and dynamic urban world. The ultimate vision is to transition path planning from a pre-mission computation to a persistent, adaptive, and resilient capability, thereby unlocking the full potential of UAV swarms for complex urban applications.

Ethical approval

This article does not contain any studies with human or animal subjects performed by any of the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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